

Research papers

Optimal configuration of battery energy storage for resilience enhancement in multi-energy-coupled active distribution networks against typhoon events

Lei Chen^{a,b,*}, Yi Zheng^{a,b}, Bude Su^{a,b}, Musi Li^{a,b,*}, Yifei Li^{a,b}, Hongkun Chen^{a,b}

^a School of Electrical Engineering and Automation, Wuhan University, Wuhan, 430072, China

^b Hubei Engineering and Technology Research Center for AC/DC Intelligent Distribution Network, School of Electrical Engineering and Automation, Wuhan University, Wuhan, 430072, China

ARTICLE INFO

Keywords:

Multi-energy coupled active distribution networks
Resilience enhancement
Optimal configuration
Battery energy storage devices
Fault recovery

ABSTRACT

With the growing integration of various energy systems, multi-energy coupled active distribution networks (MECADNs) that combine electricity, water, gas, and heat have become essential infrastructure for ensuring a reliable energy supply. Enhancing the resilience of MECADNs against typhoon disasters is therefore of great significance. This paper proposes an optimal configuration method for battery energy storage devices (BESDs) to enhance the resilience of MECADNs. Based on a dynamic model of multi-energy coupled networks, a two-layer optimization model is developed. The inner-layer focuses on the operational optimization of active distribution networks (ADNs), considering multi-energy effects, while the outer-layer targets the optimal configuration of BESDs. Additionally, a hybrid intelligent algorithm combining particle swarm optimization (PSO) and snow ablation optimizer (SAO) is designed to improve solution efficiency and accuracy. The proposed method is validated through cross-verification in small- and large-scale MECADNs based on modified IEEE 33-node and 118-node systems, integrating theoretical analysis, numerical simulation, and resilience index quantification. The main conclusions are as follows: (1) The two-layer optimization model exhibits excellent scalability across system scales, and the SAO-PSO hybrid algorithm outperforms standalone PSO/SAO in both speed and accuracy, economically reducing BESDs configuration cost in the 33-node and 118-node systems. (2) Under multi-stage typhoon fault scenarios, the method realizes simultaneous optimization of BESDs configuration and MECADNs operation, coordinating electricity-gas-water-heat subsystems to smooth load fluctuations and stabilize state parameters. (3) It significantly enhances system resilience: the average load recovery rate reaches 99.3 % in the 33-node system and rises from 78.3 % to 97.8 % in the 118-node system; resilience scores increase from 42.00 to 43.42 for small-scale MECADNs and 58.52 to 60.51 for large-scale MECADNs, with the large-scale networks showing a 40.1 % higher gain in score increment.

1. Introduction

1.1. Motivation

With the deepening integration of electricity, water, gas, and heat systems, multi-energy coupled active distribution networks (MECADNs) have evolved into core infrastructure supporting both energy security and socioeconomic stability [1,2]. Their ability to coordinate multi-energy flows enables efficient complementarity between distributed

generators (DGs) and diverse loads, thereby improving energy supply reliability [3–5]. Nevertheless, the frequency and intensity of extreme weather events, particularly typhoon disasters, have increased in recent decades, posing unprecedented threats to MECADNs [6,7].

Typhoons, characterized by strong winds, heavy rainfall, and storm surges, can trigger cascading failures across MECADNs, including multi-energy supply disturbances that easily induce voltage fluctuations in the electrical network. These high-impact, low-probability (HILP) events typically last for days, leading to extended outages of critical loads and

* Corresponding authors at: School of Electrical Engineering and Automation, Wuhan University, Wuhan, 430072, China.

E-mail addresses: chen_lei@whu.edu.cn (L. Chen), musili.manchester@gmail.com (M. Li).

<https://doi.org/10.1016/j.est.2025.120164>

Received 29 September 2025; Received in revised form 24 November 2025; Accepted 24 December 2025

Available online 8 January 2026

2352-152X/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

significant economic and social losses [8]. In this context, enhancing MECADN resilience—defined as the ability to resist, adapt to, and recover from extreme events—has become an urgent priority.

Battery energy storage devices (BESDs) are widely recognized as an adaptable and efficient approach to enhancing resilience, as they can buffer distributed generation output volatility and provide emergency power support during fault periods [9,10]. However, the optimal configuration of BESDs in MECADNs under typhoon scenarios faces unique challenges: (1) dynamic coupling among electricity, water, gas, and heat requires BESDs to be coordinated with multi-energy flows; (2) typhoon-induced multi-fault and time-varying conditions greatly increase the complexity of coordinating BESDs with other energy systems; and (3) improper sizing or siting of BESDs may lead to underutilization or insufficient resilience. These challenges highlight the necessity of a systematic optimization framework specifically tailored to MECADNs under typhoon conditions.

1.2. Literature review

To contextualize the proposed work, this study reviews relevant studies, including resilience assessment of MECADNs under extreme events, multi-energy storage technology support, as well as optimization models and algorithms for configuring BESDs to enhance resilience.

1.2.1. MECADNs resilience under extreme events

In [11], scholars develop a resilience analysis model for MECADNs and introduce a decentralized operating paradigm for real-time coordination to restore system loads, but this work focuses only on the post-disaster recovery stage and neglects the dynamic coupling of multi-energy flows during typhoon evolution. In [12], the continuous development of resilience in MECADNs in response to extreme events is analyzed, and a multi-agent, distributed operation and settlement model based on a hierarchical solution is proposed to enhance resilience, yet it fails to fully capture the multi-period fault characteristics of typhoons. In [13], a probabilistic resilience assessment framework is constructed for MECADNs, utilizing an advanced modeling approach to capture the random effects of component failures on load point connectivity. The finding demonstrates that the method enables automatic differentiation for analyzing the sensitivity of component importance measures, but it lacks integration with recovery strategies. In [14], scholars propose an optimal resilient operation model for MECADNs based on the energy-to-mobility approach. This model involves investing in and routing mobile energy storage systems, while also considering the coordinated management of various recovery actions. In [15], a robust optimization model is designed to account for the uncertainty in repair times of faulty components in MECADNs, while a decentralized synthetic fault restoration strategy is suggested to reduce interruption duration and improve resilience.

In [16], a three-phase resilience model for MECADNs is introduced, considering reliability, vulnerability, and recovery resilience. Additionally, the method incorporates initial performance, robustness, while a multi-objective fault recovery strategy implemented using the modified NSGA-II. In [17], a three-layer, data-driven resilient strategy is proposed for MECADNs. This strategy encompasses a proactive scheduling framework that identifies regions most affected by extreme events and determines the optimal location and sizing of electric vehicle charging stations to enhance resilience, but it only optimizes charging station deployment and does not involve BESD configuration for multi-energy systems. In [18], scholars regard critical load restoration as the goal of resilience, proposing an entropy-based multi-agent deep reinforcement learning approach. This method aims to maximize the use of controllable energy sources and network reconfiguration in MECADNs to enhance restoration effectiveness, yet it does not explicitly integrate BESD configuration with multi-energy dynamics.

In [19], a multi-temporal optimal power flow (OPF) model for radial networks is proposed. The model effectively incorporates the dynamic

behavior of electrical energy storage (EES) for day-ahead operation and is verified under both normal and contingent events. Note that, this work focuses on general contingent events rather than typhoon-induced multi-period dynamic fault evolution.

1.2.2. Multi-energy storage technology support for resilience enhancement

Multi-energy storage and key component optimization are increasingly recognized as important pillars for MECADN resilience, as they enrich the technical landscape of multi-energy flow balancing during extreme events. Recent studies have made significant progress in thermal energy storage (TES) and multi-energy component design, providing valuable insights for multi-energy system resilience enhancement. In [20], scholars developed a biomass-based composite phase change material (PCM) from wild daisy stems, achieving an excellent thermal buffering performance, which is critical for stabilizing thermal loads. In [21], experts conducted pore-scale numerical research on PCM microcapsule packed-bed systems, showing that the proposed approach can reduce thermocline degradation, providing insights for optimizing TES integration with MECADNs. In [22], researchers further optimized medium-low temperature packed-bed TES systems, and the structures balance economic efficiency and heat storage capacity—valuable for multi-energy load matching. Beyond TES, key multi-energy components also contribute to resilience.

In [23], a dual-flow latent heat storage system with optimized finned-tube heat exchangers was designed, thereby significantly improving charging efficiency. In [24], a hybrid printed circuit heat exchanger (PCHE) for a hydrogen refueling station was proposed, featuring a semicircle-serrated structure that reduces volume while lowering pressure drop, thereby enhancing the reliability of hydrogen supply and potentially serving as a technology to increase resilience in multi-energy systems. These studies enrich the technical foundation of multi-energy storage and component optimization, with a primary focus on thermal energy storage and single-component performance enhancement. Against this backdrop, our work focuses on the optimal configuration of BESDs in typhoon-specific scenarios, aiming to complement the multi-energy resilience research landscape by addressing the unique challenges of BESD deployment under dynamic typhoon disturbances.

1.2.3. Optimal configuration of BESDs for MECADNs

In [25], a comprehensive model based on multi-temporal optimal power flow (MTOPTF) is developed for unbalanced low-voltage (LV) networks in local energy communities (LECs), considering grid-scale batteries and addressing technical issues in islanded microgrid operational planning. In [26], a methodological framework is presented for optimizing network expansion and investments in BESDs to reduce cascading outages. The findings suggest that incorporating cascading outages into investment planning leads to different network enhancement choices; however, the framework focuses solely on the electrical network, overlooking the coupling demands of water, gas, and heat loads during typhoons. In [27], a method employing deep reinforcement learning is proposed for efficiently deploying distributed mobile energy storage to quickly restore critical loads after disasters. Case studies on 6- and 33-bus systems confirm the method's effectiveness.

In [28], the study presents a comprehensive framework for optimally allocating PV and BESDs in ADNs. The goal is to minimize energy losses and energy not served (ENS) under normal conditions, as well as power load interruptions during emergency scenarios, but resilience under typhoon scenarios is not fully addressed. In [29], a resilience-focused planning model is introduced for MECADNs to decrease load shedding during typhoons. It utilizes a tri-layer stochastic-robust optimization to determine the optimal design of energy storage systems considering uncertainties, though the computational burden remains high.

In [30], scholars introduce a modified planning approach for stationary-mobile integrated BESDs with spatial flexibility. It is designed to seamlessly switch between stationary and mobile modes, ensuring

effective operation during normal conditions and extreme events, but it does not integrate dynamic multi-energy flow characteristics into configuration decisions. In [31], a quantitative metric is proposed to evaluate resilience restoration, and emergency mobile BESDs are optimized for resilience advantages and costs through multi-agent distribution. In [32], this study introduces a novel method for determining the best installation sites and capacity distribution for BESDs. It accounts for geographical factors, safety, energy flexibility, and accessibility, but remains limited to static scenarios without multi-period disaster modeling. In [33], hybrid hydrogen systems and BESDs are optimized to enhance resilience and increase renewable energy utilization through a bi-layer decomposition algorithm. In [34], the work presents a coordinated framework for configuring soft open points (SOPs) and BESDs within MECADNs. This framework optimizes equipment capacity and operational strategies simultaneously, effectively managing uncertainties in renewable energy generation while balancing economic efficiency with grid resilience, but it focuses primarily on electrical network stability and overlooks the resilience demands of water, gas, and heat loads. In [35], the study introduces a robust multi-objective planning framework for renewable energy sources and BESDs. The model seeks to minimize total annualized costs as well as source-load mismatch, improving economic efficiency as well as operational reliability, though it emphasizes general reliability rather than resilience enhancement under typhoon conditions.

1.2.4. Research gaps

In summary, the research on enhancing the resilience of MECADNs under extreme events and optimizing the configuration of BESDs has made progress, but it still faces obvious limitations. Studies on MECADNs' resilience have explored assessment frameworks and post-fault recovery strategies, yet they either focus on static resilience evaluation or single-stage fault scenarios, failing to integrate the coupling characteristics of electricity-water-gas-heat systems into the optimal configuration of BESDs. Meanwhile, research on BESDs' optimal configuration has proposed methods such as hierarchical optimization and deep reinforcement learning, but these approaches often prioritize the electrical network alone (ignoring multi-energy synergy), and suffer from inefficiencies in solving two-layer configuration-operation models. Specifically, single-algorithm solutions can neither adapt to dynamic multi-energy coupling nor balance solution speed and global optimality, while traditional resilience-oriented configurations only reflect resilience indirectly through load interruption reduction, lacking a direct quantitative link between the configuration of BESDs and resilience score.

Regarding the research gaps, three key limitations are further clarified based on the above literature: (1) Insufficient integration of typhoon multi-period dynamics and multi-energy coupling into BESDs configuration; (2) Inefficient solution of two-layer configuration-operation models for multi-energy systems; (3) Lack of direct quantification of resilience improvement from BESDs configuration.

This paper puts forward an innovative BESDs optimal configuration method for MECADNs resilience enhancement, based on a dynamic multi-energy coupling model and a two-layer optimization framework. Meanwhile, the adoption of both the two-layer model and the SAO-PSO hybrid algorithm in this study serves a specific purpose—to address the

disconnection between multi-energy dynamics and configuration, as well as the bottleneck of solution efficiency.

1.3. Contributions of this paper

To systematically differentiate the proposed work from existing studies and highlight its innovations, a taxonomy Table 1 is shown below, covering key aspects such as research object, consideration of extreme scenarios, resilience measurement, and multi-energy coupling integration. The taxonomy table selects representative literature based on three main features: (1) focusing on the two main research themes (MECADNs resilience under extreme events and BESDs optimal configuration); (2) including various research scenarios (typhoon/general contingent events) and levels of energy coupling (single-energy/partial multi-energy/no coupling); (3) reflecting typical limitations (indirect resilience measurement, single-energy focus) that this study aims to address.

Table 1 clearly shows that our work differentiates from existing studies by integrating typhoon multi-period scenarios, direct resilience quantification, and full multi-energy coupling, addressing the key research gaps in MECADNs' resilience enhancement. The main novelty and contributions are outlined below.

- (1) Develop a dynamic coupling model of MECADNs that integrates electricity, water, gas, and heat dynamics, and embed it into a two-layer optimization framework—with the outer-layer optimizing BESDs' location and capacity, and the inner-layer optimizing ADNs' operation by considering multi-energy effects—to realize the coordination between BESDs' configuration and MECADNs' operation.
- (2) Propose a SAO-PSO hybrid intelligent solution algorithm specifically tailored for the proposed model, which integrates the local refinement capability of PSO and the global search strength of SAO. Case studies on small- and large-scale MECADNs, based on modified IEEE 33-node and 118-node systems, confirm that this algorithm outperforms standalone PSO or SAO in both solution quality and speed, effectively boosting the solving efficiency of the proposed two-layer model and breaking through its computational bottleneck.
- (3) Validate the proposed method via multi-scale case studies under multi-period typhoon fault scenarios, and quantitatively achieve a 19.5 % increase in load recovery rate—thereby providing direct evidence for the method's effectiveness and resolving the issue of inadequate direct quantification of resilience improvements.

1.4. Organization structure of this paper

The rest of this paper is structured as follows. Section 2 focuses on the dynamic modeling of MECADNs and the design of the optimal configuration model for BESDs. Section 3 addresses the model solution methods. Section 4 conducts case studies to verify the proposed approach. Section 5 summarizes the critical findings and outlines future research directions.

Table 1
Comparison of the proposed method with existing work.

Item	Research object	Extreme scenario	Resilience Quantification	Multi-energy coupling
[11]	MECADNs	Post-disaster	Indirect (load recovery)	Ignored coupling
[19]	Radial networks	General contingent events	Not involved	No
[25]	LV networks	Main grid outage	Indirect (ENS, load curtailment)	No
[26]	Radial networks	Cascading outages	Indirect (outage reduction)	No
[29]	MECADNs	General reliability scenarios	Indirect (load shedding reduction)	Partial (electricity-centric)
[34]	MECADNs	Typhoon	Indirect (grid stability)	Partial (electricity-centric)
Our work	MECADNs	Typhoon	Direct (resilience score)	Full (electricity-water-gas-heat)

2. Theoretical modeling

2.1. Model description of MECADNs

Fig. 1 illustrates the schematic of MECADNs, a complex cyber-physical system that integrates electricity, gas, water, and heat subsystems. The structure reflects the multi-energy flow coupling, with key components such as gas turbines, P2G (power-to-gas), CHP (combined heat and power), and energy storage widely recognized as foundational elements of such networks [36]. In addition, the configuration references the general layout of MECADNs in typhoon-prone coastal regions (e.g., Guangdong province of China), where multi-energy infrastructure redundancy (e.g., gas-powered backup, hydroelectric reserves) is critical for resilience against typhoon-induced outages. While not directly from a single project, it encapsulates real-world design principles for multi-energy systems in disaster-prone areas [37,38]. The power features of the ADNs can be modeled as [39]:

$$\begin{cases} P_i = V_i \sum_{j \in \text{branch}_i} V_j [G_{ij} \cos \delta_{ij} + B_{ij} \sin \delta_{ij}] \\ Q_i = V_i \sum_{j \in \text{branch}_i} V_j [G_{ij} \sin \delta_{ij} - B_{ij} \cos \delta_{ij}] \end{cases} \quad (1)$$

where P_i and Q_i denote the active and reactive power of the i -th node of the ADNs. V_i and V_j are the node voltages. G_{ij} and B_{ij} represent the conductance and susceptance of branch ij . branch_i represents the set of branches connected to the i -th node. δ_{ij} is the phase angle difference for nodes i and j .

Regarding MECADNs, the coupling equipment between the ADNs and the gas system comprises power-to-gas (P2G) units, combined heat and power (CHP) units, as well as gas turbines. In this study, the CHP's operating feature is represented as [40]:

$$\begin{cases} P_{ge} = \eta_{ge} L_{ge} \\ P_{gh} = \eta_{gh} L_{gh} \end{cases} \quad (2)$$

where P_{ge} , P_{gh} denote the CHP's electrical and thermal power outputs. L_{ge} , L_{gh} denote the gas energy converted into electricity and heat. η_{ge} and η_{gh} indicate the transformation factors. The operating characteristics of the CHP can be modeled as:

$$\begin{cases} P_{CHP} \leq P_{CHP}^{\max} - c_{v1} H_{CHP} \\ P_{CHP} \geq \max\{P_{CHP}^{\min} - c_{v2} H_{CHP}, \varphi + c_m H_{CHP}\} \\ 0 \leq H_{CHP} \leq H_{CHP}^{\max} \end{cases} \quad (3)$$

where P_{CHP} denotes the electrical output of the CHP; H_{CHP} indicates the thermal output. c_{v1} and c_{v2} show the slopes of maximum and minimum intake conditions, respectively; c_m is the slope of minimum condensing conditions; φ indicates the price of standard coal.

The power consumption process of P2G mainly occurs in the electric-to-hydrogen process, which can be described as:

$$Q_{P2G} = \eta_{P2G} L_{P2G} \quad (4)$$

where Q_{P2G} means the volume of natural gas generated; η_{P2G} denotes the gas production efficiency; and L_{P2G} shows the electrical power consumed by P2G.

In this study, the gas supply for MECADNs relies on pipeline transmission [41], with the pipeline flow model described as:

$$\begin{cases} \mathbf{A}_g \mathbf{F}_g + \mathbf{S}_g - \sum \mathbf{L}_g = 0 \\ F_{g,ij} = k_{ij} (\pi_i^2 - \pi_j^2) \quad \pi_i > \pi_j \end{cases} \quad (5)$$

where $\mathbf{A}_g$ represents the node-adjacency matrix of the natural gas network. $\mathbf{F}_g$ denotes the natural gas flow matrix in pipelines. $\mathbf{S}_g$ and $\mathbf{L}_g$ stand for the gas supply flow from gas sources and gas demand flow of loads, respectively. k_{ij} is a constant that depends on the compression characteristics, physical properties, and friction coefficient of pipeline ij —it quantifies the impact of the pipeline's inherent attributes on gas flow resistance. π_i , π_j indicate the gas pressures at natural gas nodes i and j , respectively.

For the natural gas network model, this study mainly focuses on three core components, i.e., gas turbine, P2G, and CHP, to ensure its operational feasibility and accuracy: (1) the node flow continuity equation, which guarantees that the difference between the total gas supply and total gas load at each node equals the net gas flow passing through all pipelines connected to the node; (2) the loop gas pressure equation, which describes the relationship between the pressure drop of each pipeline in a closed loop and the loop's gas flow distribution; and (3) the correlation model between node gas pressure and pipeline gas flow, which quantifies how node pressure changes impact pipeline flow

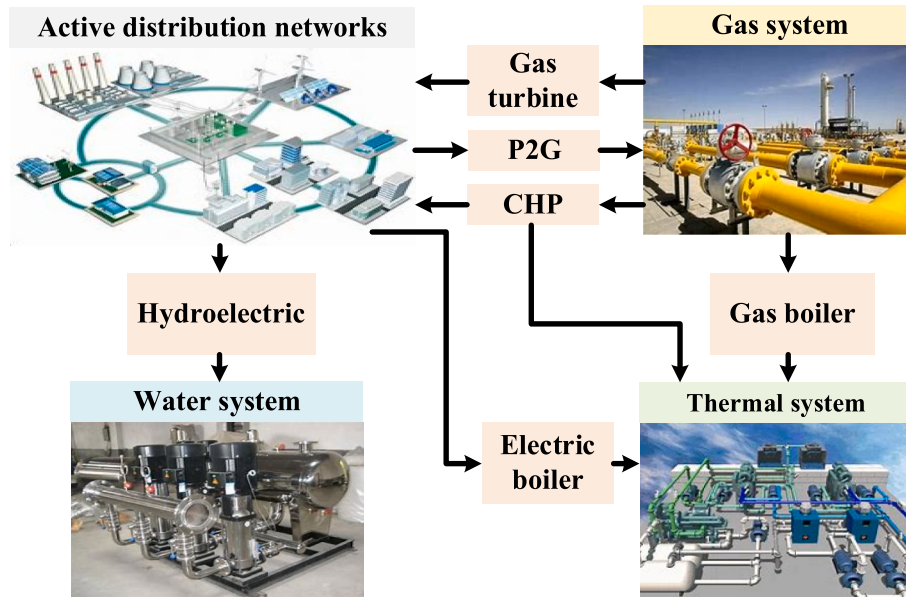


Fig. 1. Schematic diagram of MECADNs.

based on gas transportation dynamics.

Concerning the thermal system, it uses water as the working medium [42], containing the heat model as well as the hydraulic model:

$$\begin{cases} L_{hi} = C_p m_{pi} (T_{si} - T_{oi}) \\ T_{oj} = (T_{si} - T_a) e^{-\lambda l_{ij} / (C_p m_{ij})} + T_a \end{cases} \quad (6)$$

$$\mathbf{A}_w \mathbf{F}_w + \mathbf{S}_w - \sum \mathbf{L}_w = 0 \quad (7)$$

where L_{hi} stands for the thermal power at the i -th node; C_p denotes the water's specific heat capacity. T_{si} , T_{oi} indicate the water's inflow and outflow temperatures. T_a represents the environmental temperature. In addition, λ corresponds to the heat transfer factor; l_{ij} and m_{ij} are marked as the length and flow of the pipe ij . $\mathbf{A}_w$ is the thermal system's node-branch correlation matrix. $\mathbf{S}_w$ and $\mathbf{L}_w$ describe the water source as well as the water load. $\mathbf{F}_w$ is the water flow-related matrix.

The power consumed by the water pump is modeled as [43]:

$$L_w = \frac{\rho g}{\eta_w} \Delta h_w q_w \quad (8)$$

where Δh_w denotes the height of the head lift; q_w means the flow rate of the lifted water; g stands for the acceleration of gravity; and η_w shows the water pump efficiency.

The power consumed by the electric heater is deduced by [7,44]:

$$L_h = \frac{P_h}{\eta_h IR} U \quad (9)$$

where P_h indicates the heater thermal power; η_h stands for the efficiency of the heater; U corresponds to the heater voltage; I is the heater current. R denotes the electric heater resistance.

Concerning the developed water network model, this study focuses on the water transportation process in pipelines under full-flow conditions. In this state, the water network should satisfy the characteristics of pipeline equations, water pump equations, and node flow balance.

2.2. Model description of BESDs optimization

To enhance the resilience of multi-energy coupled active distribution networks (MECADNs) against typhoon events, this study introduces a two-layer optimization model based on typical fault scenarios under typhoon conditions. Notably, typical fault scenarios are chosen through a modified ORNL-PSerc-Alaska (OPA) model [40]—a choice motivated by the model's ability to integrate typhoon dynamics and component reliability characteristics. Specifically, the modified OPA model employs optimal power flow (OPF) calculations to determine information entropy and predict cascading fault paths in MECADNs, with key inputs

including the typhoon's dynamic development process and the fault probability characteristics of core components.

Fig. 2 displays the two-layer optimal configuration model of BESDs in MECADNs. The outer-layer model optimizes the BESDs' location and capacity, while the inner-layer model focuses on improving the ADNs' operation by considering multi-energy effects, thereby promoting coordination between the two-layer models.

The outer-layer model optimizes the location and capacity of BESDs from three perspectives: minimizing configuration cost, power outage loss, and node voltage deviation. The optimization variables are the location loc_{BESD_i} and capacity cap_{BESD_i} of the i -th BESD in MECADNs. The objective function f_{out} is written as:

$$\begin{aligned} \min f_{out} = & \xi_1 \sum_{i=1}^{n_{BESD}} cap_{BESD_i} pri_{BESD} \\ & + \xi_2 \sum P_{loss.outage} pri_{loss} + \xi_3 \sum U_{dis} \end{aligned} \quad (10)$$

where n_{BESD} is the allocation number of BESDs; pri_{BESD} is the unit price for BESDs; $P_{loss.outage}$ indicates the power outage loss; pri_{loss} denotes the unit power outage loss cost; U_{dis} corresponds to the voltage deviation of the ADNs. ξ_1 , ξ_2 , and ξ_3 are the weight coefficients. For the weight setting of the multi-objective function in f_{out} , this study adopts the entropy weight method for determination [45]. This method is selected because it can objectively calculate weights based on the information entropy of each objective's evaluation index—avoiding subjective bias from artificially set weights—and thereby reflect the relative importance of different objectives in the optimization process more reasonably. Specifically, in Eq. (10), the method quantifies the information entropy of three distinct objectives: (1) BESD configuration cost, (2) power outage loss, and (3) voltage deviation. By measuring the uncertainty of each objective's evaluation index, it assigns weights ξ_1 , ξ_2 , ξ_3 that reflect the inherent information characteristics of each objective. This ensures the weight assignment is purely data-driven, preventing subjective overemphasis on any single objective and enabling a holistic optimization result aligned with each goal's actual information content.

2.2.1. Constraints of the outer-layer model

1) Configuration capacity constraints

$$cap_{BESD-min} \leq cap_{BESD_i} \leq cap_{BESD-max} \quad (11)$$

where $cap_{BESD-min}$, $cap_{BESD-max}$ stand for the minimum and maximum allowable installation capacities for the i -th BESD, respectively.

2) Configuration number constraints

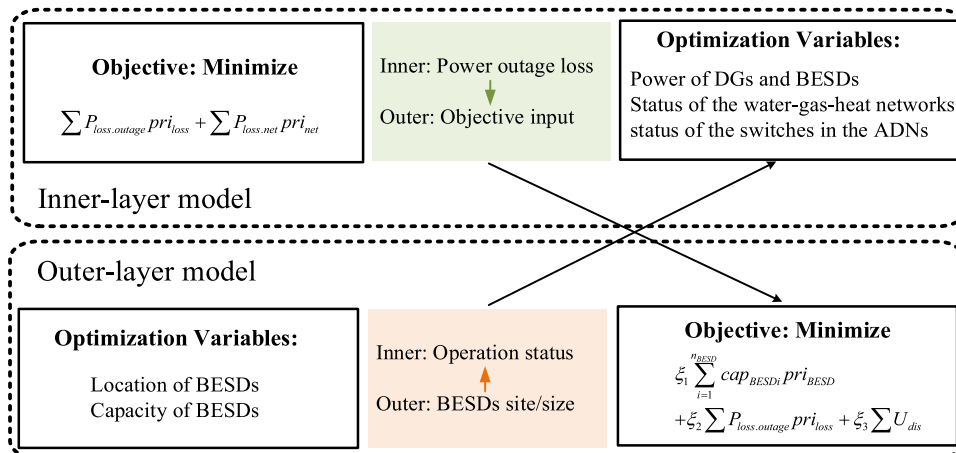


Fig. 2. Two-layer optimal configuration model of BESDs in MECADNs.

$$n_{BESD-\min} \leq n_{BESD} \leq n_{BESD-\max} \quad (12)$$

where $n_{BESD-\min}$, $n_{BESD-\max}$ denote the minimum and maximum allowable installation numbers for BESDs.

3) Configuration location constraints

$$loc_{BESD_i} \in \Omega \quad (13)$$

where Ω denotes the allowable node set for applying BESDs.

4) Node voltage deviation constraints

$$U_{dis-\min} \leq U_{dis} \leq U_{dis-\max} \quad (14)$$

where $U_{dis-\min}$, $U_{dis-\max}$ correspond to the minimum and maximum allowable node voltage deviation limits.

The inner-layer model aims to enhance fault recovery capacity by optimizing ADNs' operation while considering multi-energy effects. It considers variables such as the power adjustment of distributed generators (DGs), the power exchange between the BESDs and ADNs, the operational status of the water-gas-heat networks, as well as sectionalizing and interconnection switches within the ADNs.

The inner-layer model's objective f_{in} is to minimize the power outage loss and network loss:

$$\min f_{in} = \sum P_{loss,outage} pri_{loss} + \sum P_{loss,net} pri_{net} \quad (15)$$

where $P_{loss,net}$ indicates the network loss. pri_{net} represents the electricity price.

2.2.2. Constraints of the inner-layer model

1) DistFlow model constraints

$$\begin{cases} V_i - V_j = z_{ij} I_{ij} \quad \forall i, j \in \varepsilon \\ S_{ij} = V_i I_{ij}^* \quad \forall i, j \in \varepsilon \\ \sum_{k \rightarrow j} S_{jk} = \sum_{i \rightarrow j} (S_{ij} - z_{ij} |I_{ij}|^2) + S_j \quad \forall i, j, k \in N \\ S_j = S_{DG,j} + S_{BESD,j} + S_{COin,j} - S_{load,j} - S_{COout,j} \quad \forall j \in N \end{cases} \quad (16)$$

where S_{ij} is the power flowing from node i towards node j ; S_j stands for the power at node j ; $S_{DG,j}$ is the DG power coupled to node j ; $S_{BESD,j}$ denotes the exchange power of the BESDs coupled to node j ; $S_{COin,j}$ is the power injected into the ADNs from the water-gas-heat networks via node j ; $S_{COout,j}$ corresponds to the power output from the ADNs to the water-gas-heat networks via node j .

2) State of charge (SoC) constraints

$$\begin{cases} SoC_{i,k+1} = SoC_{i,k} + \eta_{chi} \frac{P_{chi}}{cap_{BESD_i}} \Delta t - \frac{P_{disi}}{\eta_{disi} cap_{BESD_i}} \Delta t \\ SoC_{i-\min} \leq SoC_i \leq SoC_{i-\max} \end{cases} \quad (17)$$

where $SoC_{i,k}$ is the charging state of the i th BESD in the k th calculation period. η_{chi} and η_{disi} denote the charging and discharging efficiencies. Based on [46], this work normalizes the energy change with the rated capacity, and the SoC calculation now maintains dimensionless consistency (ranging from 0 to 1), aligning with its physical definition. $SoC_{i-\max}$ as well as $SoC_{i-\min}$ stand for the maximum and minimum values of SoC regarding BESDs.

3) DG power constraints

$$\begin{cases} P_{DGj-\min} \leq P_{DGj} \leq P_{DGj-\max} \\ Q_{DGj-\min} \leq Q_{DGj} \leq Q_{DGj-\max} \end{cases} \quad (18)$$

where P_{DGj} and Q_{DGj} represent the active and reactive power outputs of the DG unit at node j ; $P_{DGj-\min}$, $P_{DGj-\max}$, $Q_{DGj-\min}$, and $Q_{DGj-\max}$ stand for the minimum and maximum allowable power limits.

4) Operation constraints for the water-gas-heat networks

The operation of the natural gas network mainly involves meeting the power flow constraints, gas load constraints, node gas pressure limits, and pipeline transmission capacity constraints of the gas network, which can be expressed as:

$$\begin{cases} \pi_{\min} \leq \pi \leq \pi_{\max} \\ F_{g-\min} \leq F_g \leq F_{g-\max} \end{cases} \quad (19)$$

where π , $\pi_{\max}$, and $\pi_{\min}$ represent the node gas pressure and its maximum and minimum limits; $F_{g-\max}$ and $F_{g-\min}$ denote the maximum and minimum gas flows of the pipelines.

The operating constraints of the heat and water networks can be expressed as:

$$\begin{cases} F_{w-\min} \leq F_w \leq F_{w-\max} \\ T_{\min} \leq T_{si} \leq T_{\max} \\ T_{\min} \leq T_{oi} \leq T_{\max} \end{cases} \quad (20)$$

where $F_{w-\max}$ and $F_{w-\min}$ indicate the maximum and minimum water flows of the pipelines. $T_{\min}$ and $T_{\max}$ denote the lower and upper limits of the water temperature.

5) Constraints on radial topology of the ADNs

$$\sum y_{ij} = N - 1 \quad \forall (i, j) \in \varepsilon \quad (21)$$

$$\alpha_{ij} + \alpha_{ji} = y_{ij} \quad \forall (i, j) \in \varepsilon \quad (22)$$

$$\begin{cases} \sum_{j \in N} \alpha_{ij} = 0 & i = 1 \\ \sum_{j \in N} \alpha_{ij} = 1 & i = 2, 3, \dots, N \end{cases} \quad (23)$$

where ε corresponds to the set of all branches of the ADNs and Eq. (21) is the radiality constraint [47]. Therefore, the total number of branches in operation must equal the total number of nodes minus one. α_{ij} is a relational variable characterizing between node i and node j on branch ij . If the power flows from node i to node j , then $\alpha_{ij} = 1$, and vice $\alpha_{ij} = 0$.

3. Model solution method

This study constructs a two-layer optimal configuration model for BESDs in the MECADNs. The outer-layer model's objective is based on the optimal operation scheme derived from addressing the inner-layer model, establishing a closed-loop interaction between the two layers. To balance solution efficiency and optimality, a computationally efficient heuristic algorithm is adopted to handle the outer-layer model, while a mathematical optimization method addresses the inner-layer model, exploiting the mathematical method's advantage in ensuring solution optimality for operation-centric problems.

3.1. Outer-layer model solving algorithm

The outer-layer model's objective and constraints are coupled with the operation results of the inner-layer model, resulting in a complex optimization problem that pure mathematical methods cannot directly solve due to potential non-convexity and high dimensionality. To overcome this, this section employs a hybrid optimization algorithm that integrates the strengths of two intelligent algorithms, i.e., particle swarm optimization (PSO) and snow ablation optimizer (SAO).

3.1.1. PSO algorithm

The PSO algorithm is inspired by the foraging behavior of bird flocks, aiming to seek the global optimal solution by emulating the information-sharing and population cooperation mechanisms among individuals [48,49]. In PSO, each candidate solution is regarded as a “particle” that traverses the solution space freely. The algorithm realizes iterative evolution through continuous updates of particles' velocities and positions, where the update rules are formulated as:

$$\begin{cases} v_i^{k+1} = v_i^k + c_1 r_1 (p_{best,i}^k - x_i^k) + c_2 r_2 (g_{best}^k - x_i^k) \\ x_i^{k+1} = x_i^k + v_i^{k+1} \end{cases} \quad (24)$$

where v stands for the velocity of the particle; c_1 and c_2 are the preset hyperparameters of the algorithm, representing the self-learning factor as well as social learning factor. r_1 as well as r_2 are random numbers that are updated in each iteration, and the range of change is $[0, 1]$; $p_{best,i}^k$ is the personal best position of the i -th particle in the k iterations; g_{best}^k denotes the global best position among all particles; x stands for the coordinates of the i -th particle, as a candidate solution of the optimization problem.

3.1.2. SAO algorithm

The SAO algorithm relies on the Brownian motion model to mimic the diffusion behavior of vapor within spatial domains [50,51]. The SAO not only boosts global search capability but also robustly mitigates the risk of premature convergence to local optima—an advantage over many traditional metaheuristic algorithms, which often suffer from premature convergence. Typically, the SAO divides the snow molecule population into two groups, i.e., the sub-populations P_a as well as P_b . The number of snow molecules can be controlled by:

$$N_a(t+1) = \begin{cases} N_a(t) + 1 & N_a(t) < N \\ N & N_a(t) = N \end{cases} \quad (25)$$

$$N_b(t+1) = N_b(t) - 1 \quad (26)$$

where $N_a(t)$ and $N_b(t)$ stand for the numbers of snow molecules in P_a and P_b . $N_a(0) = N_b(0) = N/2$; N corresponds to the total snow molecules.

For P_a , it searches for the optimal solution by simulating the Brownian motion of water vapor and fine-tuning the parameters:

$$\mathbf{Z} = \mathbf{Z}_{min} + \mathbf{Z}_{rand}(\mathbf{Z}_{max} - \mathbf{Z}_{min}) = \begin{bmatrix} Z_{1,1} & \cdots & Z_{1,dim} \\ \vdots & \ddots & \vdots \\ Z_{N,1} & \cdots & Z_{N,dim} \end{bmatrix} \quad (27)$$

$$\begin{aligned} Z_{ia}(t+1) &= Elite(t) + BM_i(t) \otimes \\ &\{ [\theta_1 \times (G(t) - Z_{ia}(t))] + (1 - \theta_1) \times [\bar{Z}(t) - Z_{ia}(t)] \} \end{aligned} \quad (28)$$

where $\mathbf{Z}$ stands for the position information matrix with the limits of $\mathbf{Z}_{min}$ and $\mathbf{Z}_{max}$. $\mathbf{Z}_{rand}$ indicates a random number matrix. $Z_{ia}(t+1)$ stands for the position of the i -th molecule in P_a at $t+1$. $Elite(t)$ means randomly selected among $G(t)$, $Z_{second}(t)$, $Z_{third}(t)$, and $Z_c(t)$. $BM_i(t)$ is defined as the Gaussian distribution. $G(t)$ corresponds to the molecule's optimal position, $Z_{second}(t)$ and $Z_{third}(t)$ are the second and third optimal positions, respectively. $Z_c(t)$ represents the average position.

For P_b , it searches for the optimal solution by simulating the snow-melt exploration and updating the parameters:

$$\mathbf{M} = \left(0.35 + 0.25 \times \frac{e^{\frac{t}{t_{max}} - 1}}{e - 1} \right) \times (T - T_1) \quad (29)$$

$$\begin{aligned} Z_{ib}(t+1) &= \mathbf{M} \times G(t) + BM_i(t) \otimes \\ &\{ [\theta_2 \times (G(t) - Z_{ib}(t))] + (1 - \theta_2) \times [\bar{Z}(t) - Z_{ib}(t)] \} \end{aligned} \quad (30)$$

where $\mathbf{M}$ is the snow ablation rate. T and T_1 stand for the current average

temperature and initial temperature, respectively. θ_1 and θ_2 are random constants between $[-1, 1]$. $Z_{ib}(t+1)$ indicates the position of the i -th molecule in P_b at $t+1$.

3.1.3. SAO-PSO hybrid algorithm

The PSO algorithm, despite its simple search mechanism and random particle distribution, has inherent limitations in complex optimization situations. The PSO primarily depends on two main sources of information during iteration: the personal best position and the global best position. This over-reliance on these sources often fails to utilize sub-optimal solutions, the characteristics of population distribution, and other useful search clues. As a result, it can lead to premature convergence on local optima and reduce the accuracy of the final solutions. To address these limitations, the SAO-PSO hybrid algorithm is developed by combining the optimization process of the SAO algorithm. In fact, the SAO performs well in global exploration through Brownian motion-inspired diffusion and multi-source information fusion. By integrating these strengths into PSO, the hybrid algorithm aims to improve convergence accuracy through detailed optimal information exchange and reduce the risk of getting stuck in local optima with its robust global search abilities.

To integrate the SAO's advantages into the PSO, this work expands the guidance information set beyond $p_{best,i}$ and g_{best} (defined in Eq. (24)) by introducing: (1) G_{second}^k and G_{third}^k , i.e., the second-best and third-best positions in the swarm in the k iterations; (2) $\bar{x}^k$, i.e., the average position of all particles in the swarm (analogous to the SAO). Hence, an elite selection mechanism is applied to randomly pick guidance from top-tier positions:

$$E_i^k = \text{randselect} \{ g_{best}^k, G_{second}^k, G_{third}^k \} \quad (31)$$

where $\text{randselect} \{ \cdot \}$ denotes a random selecting operation.

Building on the standard PSO velocity update in Eq. (24), the SAO-PSO hybrid algorithm modifies the velocity rule to incorporate the SAO's multi-source guidance:

$$\begin{cases} v_i^{k+1} = v_i^k + c_1 r_1 (p_{best,i}^k - x_i^k) + c_2 r_2 (E_i^k - x_i^k) + c_3 r_3 (\bar{x}^k - x_i^k) \\ x_i^{k+1} = x_i^k + v_i^{k+1} \end{cases} \quad (32)$$

where v_i^{k+1} , v_i^k , x_i^{k+1} , and x_i^k are the velocities and positions of the i -th particle at the $k+1$ and k iterations, respectively. c_3 is a new acceleration coefficient governing population average guidance. r_3 is within $[0, 1]$ and is an additional random number to enhance exploration stochasticity.

In this context, Fig. 3 shows the flowchart of the improved SAO-PSO hybrid algorithm. The process includes: (1) Initializing all particles and algorithm parameters; (2) Conducting Brownian motion and multi-source fitness fusion to explore promising regions; (3) Updating sub-groups via the SAO and particle positions via the PSO; (4) Checking convergence criteria to determine if the algorithm terminates.

3.2. Inner-layer model solving algorithm

The core challenge in solving the inner-layer model arises from its multivariate, highly coupled, and nonlinear constraints, which originate from: (1) nonlinear DistFlow equations (Eq. (16)) with quadratic terms (e.g., $z_{ij}|I_{ij}|^2$); (2) nonlinear SoC evolution for BESDs (Eq. (17)) involving charging/discharging efficiency factors (η_{ch} , η_{dis}); (3) linear bounds for DG active/reactive power (Eq. (18)); and (4) operational constraints for water-gas-heat networks (Eqs. (19)–(20)) with discrete or state-dependent behaviors. These nonlinear and non-convex structures not only increase computational complexity but also render conventional convex optimization algorithms ineffective, as they lack mechanisms to handle non-convex feasible regions or discrete decision spaces. Thus, tailored reformulations into tractable convex forms are required, which

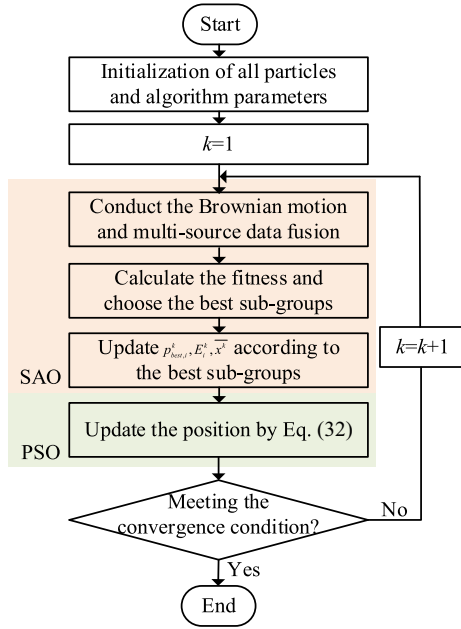


Fig. 3. Flowchart of the improved SAO-PSO hybrid algorithm.

is achieved here via second-order cone relaxation (SOCR) and the Big-M linearization technique [52–54]. Specifically, the Big-M technique is employed to linearize discrete or state-dependent operational constraints (e.g., binary decisions in multi-energy network operations) by introducing auxiliary binary variables and linear inequalities scaled by a sufficiently large parameter M . This converts non-linear or discrete logical relationships into tractable linear forms, enabling the inner-layer model to be solved by commercial linear programming solvers and ensuring both computational efficiency and solution feasibility.

To address the nonlinear DistFlow constraints in Eq. (16), we use the SOCR to relax the term $z_{ij}|I_{ij}|^2$ into a convex second-order cone (SOC) constraint:

$$\left\| \text{Re}(S_{ij}), \text{Im}(S_{ij}), \frac{|V_i|^2 - |V_j|^2}{2} \right\|_2 \leq \frac{|V_i|^2 + |V_j|^2}{2} - \text{Re}(S_{ij}) \quad (33)$$

This relaxation is theoretically justified by preserving the radial distribution network's power balance physics while converting the non-convex quadratic term into a convex conic form. Notably, it retains tightness for typical distribution network operating conditions, ensuring the relaxed solution closely approximates the original problem's optimal point. This embedding enables efficient solutions within the second-order cone programming (SOCP) framework, leveraging its polynomial-time solvability for convex cases.

For the BESD SoC dynamics (Eq. (17)), we introduce binary variables $y_{ch,i}$ and $y_{dis,i}$ to denote charging and discharging states, respectively, and a third binary variable $y_{idle,i}$ to represent the IDLE mode, ensuring mutual exclusivity among charging, discharging, and IDLE states. Further, we implement the Big-M for linearizing these nonlinearities:

$$\begin{cases} \text{SoC}_{i,k+1} = \text{SoC}_{i,k} + \eta_{ch} \frac{P_{ch}}{\text{cap}_{\text{BESD}i}} \Delta t - \frac{P_{dis}}{\eta_{dis} \text{cap}_{\text{BESD}i}} \Delta t \\ \quad + M(1 - y_{ch,i} - y_{idle,i}) - My_{dis,i} \\ y_{ch,i} + y_{dis,i} + y_{idle,i} = 1 \\ \text{SoC}_{i-\min} \leq \text{SoC}_i \leq \text{SoC}_{i-\max} \end{cases} \quad (34)$$

where M is a sufficiently large constant that enforces mutual exclusivity between charging and discharging states. Based on [55], $y_{idle,i} = 1$ indicates the i th BESD is in IDLE mode (neither charging nor discharging),

so $P_{ch} = P_{dis} = 0$, and $\text{SoC}_{i,k+1} = \text{SoC}_k$. When $y_{ch,i} = 1$ (charging active), the term $M(1 - y_{ch,i})$ vanishes, while $-My_{dis,i}$ relaxes the discharging term (since $y_{dis,i} = 0$); the reverse holds for discharging active. This linearization preserves operational logic while eliminating efficiency-induced nonlinearities, ensuring compatibility with mixed-integer optimization frameworks.

Beyond these core nonlinear components, DG power constraints (Eq. (18), $P_{DGj-\min} \leq P_{DGj} \leq P_{DGj-\max}$ and $Q_{DGj-\min} \leq Q_{DGj} \leq Q_{DGj-\max}$) are directly incorporated as linear bounds in the optimization model. For water-gas-heat network constraints (Eqs. (19)–(20)), mild nonlinearities are retained as linear inequalities, while discrete operational states are linearized via Big-M, ensuring seamless integration with the overall optimization structure.

As a result, the inner problem is transformed into a mixed-integer second-order cone programming (MISOCP) model, a well-established optimization paradigm with proven solvability using modern solvers (e.g., CPLEX) [56]. Compared with directly tackling the original nonlinear non-convex problem, this approach significantly enhances computational efficiency, solution feasibility, and scalability, while maintaining sufficient accuracy to capture the critical physical and operational characteristics of the MECADNs.

4. Case study

To assess the effectiveness and flexibility of the proposed method, different sizes of MECADNs are modeled, while various intelligent algorithms are evaluated for performance comparison. The tests are conducted in MATLAB 2020a on a personal computer with an Intel Core i7–9700 CPU running at 3.00 GHz and 16 GB of RAM.

Prior to conducting the case simulations, a dedicated validation of the test models is performed to ensure the reliability. Following the method proposed by [57], we compare the power flow results of the base case scenarios (IEEE 33-bus and 118-bus networks) with the quadratically constrained programming (QCP) model with McCormick relaxation technique. Key system parameters are strictly aligned for fair comparison: the slack bus voltage is set to 1 p.u. for both test systems; the IEEE 33-bus system adopts the nodal active/reactive power demands from the reference (total active load: 3.715 MW, total reactive load: 2.300 MVar), and the IEEE 118-bus system uses standard load parameters; node voltage limits are unified to 0.9–1.1 p.u. For the IEEE 33-bus system, the minimum voltage of our test model and the QCP model is 0.93 p.u. and 0.913 p.u. (difference: 0.017 p.u., within the acceptable range), and the total active power loss is 202.12 kW (deviation: 0.02 % vs the QCP model, 0.02 % vs the MATPOWER reference). For the IEEE 118-bus system, nodal voltage deviations are within 0.02 p.u., and power loss deviations are within 1 %, confirming the model's ability to reliably capture fundamental power-flow characteristics.

4.1. MECADNs model based on 33-node system

Fig. 4 displays the MECADNs model, which is based on the modified IEEE 33-node system [58], 9-node gas system [59], and the modified 32-node thermal system [60]. To clarify the connection logic and avoid visual ambiguity, all nodes are labeled with explicit network affiliations, and coupling interfaces are highlighted with distinct line styles and node markers. The detailed structure is described as follows: (1) Coupling between the water network and ADNs. The water network's node 11 is directly connected to the ADNs' node 12 via an electric pump (EP). In the diagram, these two nodes are arranged in the same regional layout, with a solid line indicating the connection path. (2) Coupling between the district heating system (DHS) and ADNs. The DHS's node 29 is coupled to the ADNs' node 27 via an electric heater (EH), which serves as the core energy-conversion equipment. (3) Coupling between the gas system and ADNs. The gas system's pipeline segments 4–8 are equipped with an electric compressor (ECO), which is supplied with electrical energy from the ADNs' node 7.

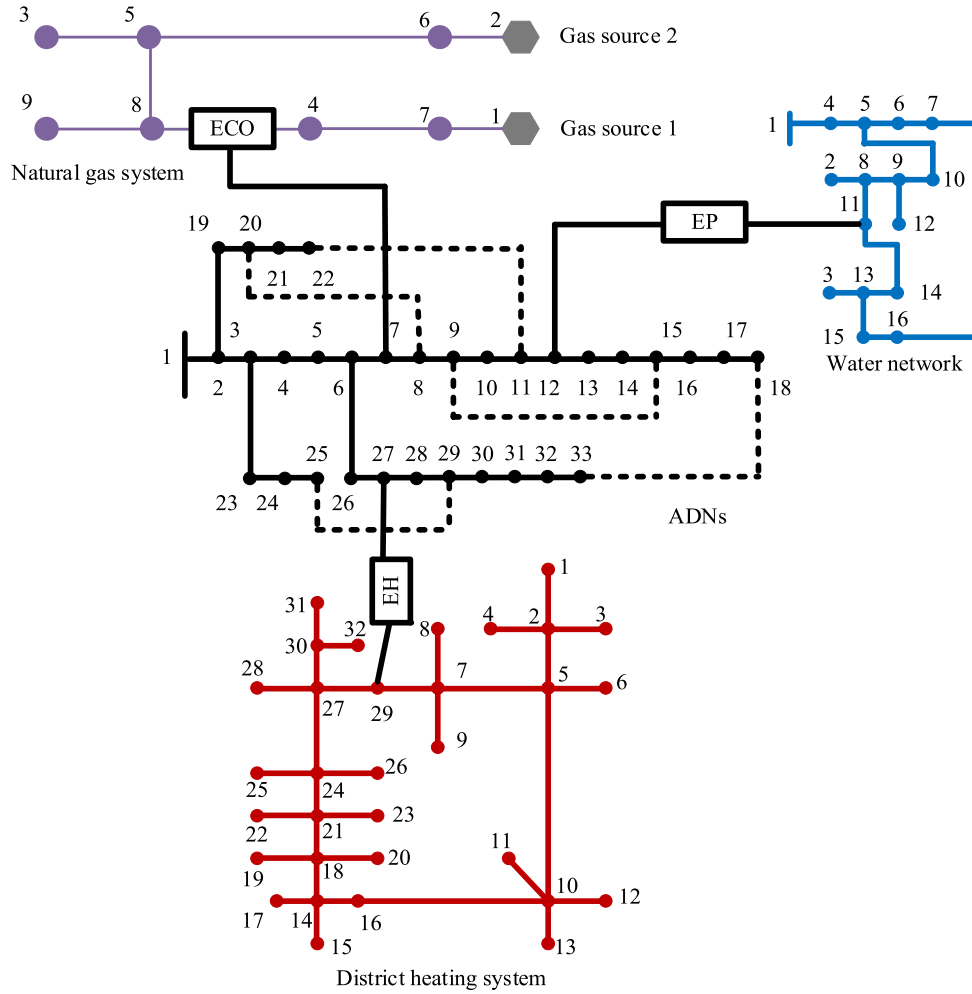


Fig. 4. MECADNs model based on the modified IEEE 33-node system.

This study adopts a data-driven, multi-dimensional simulation approach based on the Batts model to simulate a typhoon attack. The spatiotemporal impact of typhoon events is simulated by combining the typhoon's moving track and intensity attenuation law, and by calculating wind speed and wind load distribution across the MECADNs coverage area. This distribution directly determines the failure probability or performance degradation of core equipment. In addition, by integrating this wind field with the strength characteristics of core equipment (e.g., overhead lines and towers), the time-varying failure probability of distribution branches in the MECADNs is derived, exhibiting a trend where a single component's failure rate first increases and then decreases as the typhoon moves, and the component with the highest failure rate shifts with the typhoon's position. This aligns with real-world behavior, where components in regions with higher wind speeds have proportionally greater failure probabilities.

Based on the OPA model in [40], we analyze typical fault scenarios in MECADNs by incorporating the spatiotemporal variation law of typhoon events. In this study, the OPA model adopts a double-layer interactive logic. The upper-layer OPA model takes typhoon spatiotemporal data (track, intensity, and duration simulated by the Batts model) as input, first calculating the time-varying failure probability, which combines structural failure risk (e.g., from wind field) and operational failure risk (e.g., from power flow transgression). Then, it updates the fault scenario based on the combined failure probability, forming the initial cascading fault path framework. The lower-layer OPA model solves the optimal power flow (OPF) for the MECADNs, accounting for multi-energy coupling constraints. The obtained operational data (e.g., power flow

distribution, node voltages) is fed back to the upper-layer model to adjust fault constraints, enabling dynamic iteration of the fault path with typhoon movement. To generate typical fault scenarios, we first simulate 500 initial scenarios by varying typhoon parameters and component strengths. Then, we use information entropy (calculated from component failure probabilities) to screen scenarios with E in the range 60–120 (covering 90 % of high-probability cases) and retain them. In this regard, three typical fault scenarios are identified, which involve the branches 5–6 & 28–29, 15–16 & 30–31, and 14–15 & 31–32, with corresponding information entropy values of 62.3149, 63.1034, and 62.6261, respectively. Taking these typical fault scenarios as benchmark disturbance conditions, we further investigate the optimal configuration problem of BESDs in MECADNs, focusing on optimizing the site and size of BESDs to enhance the system's resilience.

To quantitatively assess the performance of different optimization algorithms in dealing with the proposed model, three test cases are constructed, where distinct algorithms solve the outer-layer model while the inner-layer model is uniformly solved via CPLEX + YALMIP:

Case 1. Solve the outer-layer model with the PSO algorithm.

Case 2. Solve the outer-layer model using the SAO algorithm.

Case 3. Solve the outer-layer model by the SAO-PSO hybrid algorithm.

Fig. 5 depicts the convergence behavior of the objective function versus iterations for the three cases. For the PSO, it converges rapidly in the early iteration stage (within 5 iterations) but then becomes trapped in local optimality due to premature convergence, a common limitation

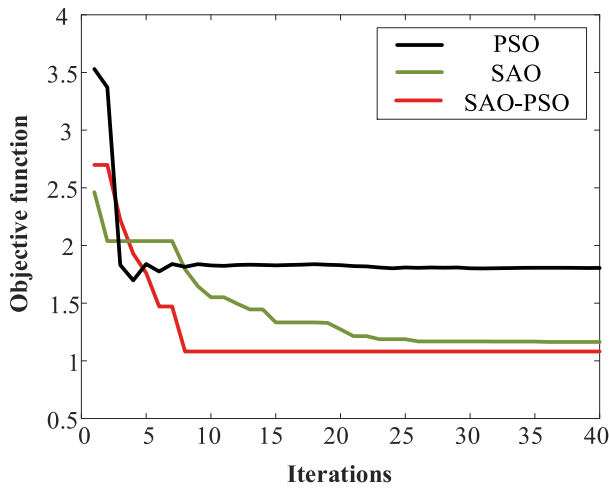


Fig. 5. Performance comparison of different algorithms to solve the proposed model.

of PSO in complex multi-energy optimization problems, ultimately stabilizing at a relatively high objective function value of ~ 1.8 . For the SAO, although it has a slow convergence rate in the initial phase, it can escape local convergence during the iterative process and converges to an objective function value of approximately 1.2 at the 26th iteration. In contrast, the SAO-PSO hybrid algorithm exhibits outstanding performance, converging to a lower objective function value (near 1.1) at the 9th iteration, thereby achieving both faster convergence speed and higher solution accuracy compared to the PSO and SAO algorithms.

Table 2 presents the optimal configuration results of the BESDs in the MECADNs model under various scenarios, with distinct characteristics in installation sites and capacity scales across the three algorithms.

In this work, the economic parameters during the optimization are anchored in industry standards and [61]. For the lithium-ion battery cost, we set it at \$100/kWh, based on [61], and converted to 710 k ¥/MWh (or 0.71 k ¥/kWh) using the exchange rate of 1 USD $\approx$ 7.1 CNY. Besides, the unit cost of the outage loads is \$167/kWh, and the unit cost of the network losses is \$3/kWh. Herein, the continuous battery sizes in Table 2 are derived from the continuous optimization model, which is designed to avoid limiting the algorithm's search space and ensure precise matching with multi-energy system dynamics. In practical engineering, these continuous values can be rounded to the nearest commercial modular capacity without affecting the optimization effect, balancing theoretical optimality and engineering feasibility.

To further emphasize the value of the proposed optimization framework, we introduce a rule-of-thumb baseline for BESD allocation, defined with conservative engineering assumptions to highlight the superiority of our optimized solution. According to this baseline, the BESD configuration is expected to meet 80 % of the ADNs' power load for 12 h with a 1.3 safety margin, accounting for extreme typhoon events and energy loss uncertainties. For the 33-node system, the required total

capacity for this baseline is calculated as $3.715 \text{ MW} \times 80 \% \times 12 \text{ h} \times 1.3 = 46.36 \text{ MWh}$. This baseline, though representative of empirical engineering practices, is inherently redundant as it neglects node-specific criticality and multi-energy coupling effects.

For the PSO algorithm (Case 1), BESDs are configured at sites 1, 15, and 31, with capacities clustered in the range of 14.52–14.98 MWh. Such a configuration features large and homogeneous capacity allocations across relatively concentrated sites. This outcome reflects the PSO's inherent limitation of premature convergence, resulting in redundant capacity and thereby increasing the total cost. In contrast, the SAO algorithm (Case 2) selects sites 17, 19, and 23, with capacities ranging from 4.32 MWh to 10.24 MWh. Here, the sites are more dispersed, and the capacity scales are more diversified. This demonstrates the SAO can explore more flexible combinations of sites and capacities for BESDs, reducing unnecessary capacity redundancy and achieving a cost savings of 16,287.4 k ¥ compared to the PSO algorithm.

It should be pointed out that, the capacity results in Case 1 (PSO algorithm) are not the global optimal solution but a consequence of PSO's premature convergence, as clearly illustrated in Fig. 5, where PSO stabilizes at a relatively high objective function value. This local optimality leads to the seemingly "excessive" capacity in Case 1, which is not designed for daily self-sustainability but rather an artifact of suboptimal algorithmic performance. The distinction between these algorithmic outcomes is critical to understanding that the capacity configuration in this study is rooted in resilience optimization for extreme events, not daily energy autonomy.

Notably, the proposed SAO-PSO hybrid algorithm (Case 3) achieves the most refined balance: it configures BESDs at sites 3, 22, and 32, with capacities distributed between 4.32 MWh and 10.97 MWh. By integrating the PSO's rapid early-stage convergence and the SAO's robust global search ability, the hybrid algorithm identifies the most demand-adaptive site-capacity combinations for configuring BESDs in the MECADNs, ultimately achieving the minimal total cost while ensuring configuration rationality. In direct contrast to the 46.36 MWh baseline, Case 3's total BESD capacity of 21.23 MWh demonstrates a 54.2 % reduction while maintaining resilience against typhoon events, underscoring the framework's superiority in balancing cost and resilience.

As the inner-layer model aims at enhancing fault recovery by optimizing the operation of ADNs while considering multi-energy effects, its impact on water-gas-heat systems is shown in Fig. 6 through both quantitative load and pressure changes, as well as qualitative operational trend differences between cases with and without the inner-layer model.

For the water system in Fig. 6(a), without the inner-layer model, the water load fluctuates more sharply, peaking at approximately 40,000 L at node 9. With the inner-layer model, the curve becomes smoother, and the peak drops to approximately 38,000 L, indicating a more even distribution of water demand. For the heat system in Fig. 6(b), the peak load at key nodes (e.g., node 28) decreases from approximately 0.5 MW to 0.4 MW with the inner-layer model, resulting in a more evenly distributed load curve across nodes. Examining the gas system in Fig. 6(c), sharp load peaks—reaching approximately 80 m³ at node 7 and 100 m³ at node 9 without the inner-layer model—are notably reduced to around 60 m³ when the model is included, indicating a decrease in sudden gas demand spikes. Gas pressure, as shown in Fig. 6(d), also improves: without the inner-layer model, it fluctuates drastically, dropping to approximately 0.9 p.u. at nodes 4 and 8; with the inner-layer model, stability improves, remaining above approximately 0.92 p.u. at most nodes.

These results collectively demonstrate that the inner-layer model enables the coordinated optimization of water-gas-heat operations—it not only smooths load and pressure fluctuations but also reduces energy waste, thereby facilitating multi-energy collaborative operation under faults.

To further verify the effects of BESDs configuration on the MECADNs under the aforementioned typhoon fault events, the multi-period

Table 2

Optimal configuration results of the BESDs in the MECADNs model based on a 33-node system.

Case	Site	Capacity/MWh	Cost/k ¥	Total cost/k ¥
1	1	14.52	10,309.2	31,360.7
	15	14.67	10,415.7	
	31	14.98	10,635.8	
2	17	10.24	7270.4	15,982.1
	19	6.72	4771.2	
	23	5.55	3940.5	
3	3	4.32	3067.2	15,073.3
	22	5.94	4217.4	
	32	10.97	7788.7	

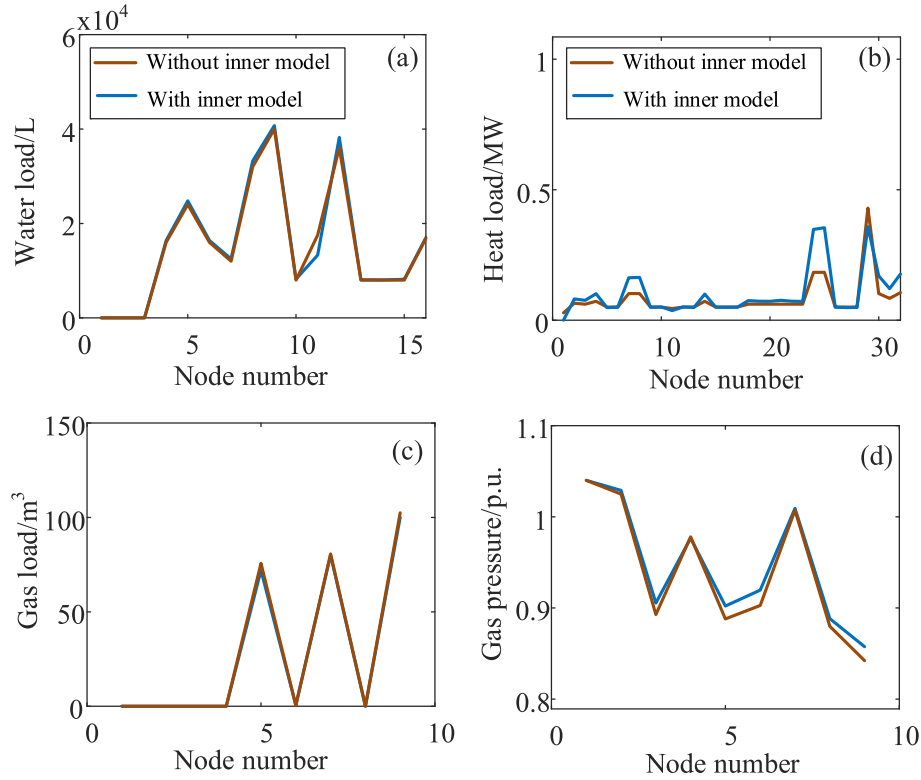


Fig. 6. The influence of the inner-layer model of BESDs configuration on the water-gas-heat systems.

typhoon fault scenarios (with three stages determining the fault evolution path) are defined in Fig. 7, and four schemes are compared as follows.

Scenario I: Without the configuration of BESDs.

Scenario II: Optimal configuration of BESDs based on the PSO algorithm.

Scenario III: Optimal configuration of BESDs based on the SAO algorithm.

Scenario IV: Optimal configuration of BESDs based on the SAO-PSO hybrid algorithm.

Fig. 8 depicts the node voltage profiles of the ADNs within the MECADNs across the three stages of multi-period typhoon fault scenarios defined in Fig. 7. Here, Scenario I represents the case without the configuration of BESDs, Scenario II implements the optimal configuration of BESDs based on the PSO algorithm, Scenario III adopts the optimal configuration of BESDs based on the SAO algorithm, and Scenario IV utilizes the optimal configuration of BESDs based on the SAO-PSO hybrid algorithm. In all stages, Scenario IV's voltage curve remains closest to the nominal 1 p.u., showing the least voltage fluctuations and dips—for example, in stage 1 (Fig. 8(a)), Scenario I experiences notable voltage drops (e.g., near node 25, voltage drops below 0.95 p.u.), while Scenario IV maintains voltages within a narrow range around 1 p.u. Similar patterns occur in stage 2 (Fig. 8(b)) and stage 3 (Fig. 8(c)), where Scenario IV's voltage curve is also smoother and nearer to 1 p.u., indicating a stronger resistance to voltage deviations caused by faults.

The voltage variations across scenarios are justified by the BESDs configuration strategy and algorithm performance: Scenario I lacks BESDs to buffer power imbalances, leading to significant voltage deviations under typhoon faults; Scenarios II–IV incorporate BESDs, but the SAO-PSO hybrid algorithm in Scenario IV achieves better coordination between BESDs' location and capacity, thus more effectively stabilizing node voltages. Quantitatively, after averaging the voltage deviations across all stages, the average deviations for Scenarios I–IV are approximately 1.531 %, 1.433 %, 1.381 %, and 1.225 %, respectively.

This confirms that Scenario IV achieves the most effective node-voltage optimization.

Fig. 9 illustrates the fault recovery performance of ADNs in the MECADNs under multi-period typhoon fault scenarios, presenting load recovery rate heatmaps for three stages across four schemes. Quantitatively, the average load recovery rates across the three stages are 88.6 % (Scenario I), 98.4 % (Scenario II), 96.1 % (Scenario III), and 99.3 % (Scenario IV), confirming Scenario IV's dominant recovery performance. Moreover, in stages 2 and 3 of Scenario IV, nearly all nodes achieve 100 % load recovery (as indicated by the dense dark blue in the heatmaps), whereas in stage 1, the recovery rate still reaches 97.8 % (with only minor light patches).

This robust recovery performance aligns with prior findings that Scenario IV also attains the lowest energy storage configuration cost. The variation in recovery rates across stages (e.g., slightly lower recovery in stage 1 vs. stages 2–3 for Scenario IV) reflects how typhoon fault locations and evolution impact system resilience, yet the SAO-PSO hybrid algorithm minimizes such impacts most effectively. Collectively, these results demonstrate that the proposed method simultaneously optimizes the configuration of BESDs for economic efficiency and maximizes load recovery during faults, achieving a dual benefit of cost savings and enhanced system resilience.

To directly quantify the resilience improvement brought by BESDs configuration, this work adopts a comprehensive resilience assessment framework by referring to [62], and Fig. 10 illustrates the structure of this resilience index system for the MECADNs. From Fig. 10, the resilience assessment system classifies indices into static indicators (SI) and dynamic indicators (DI) for each subsystem, including ADNs, natural gas system (NGS), water network (WN), and district heating system (DHS). For each subsystem, both SI and DI contain 4 specific metrics, which correspond to *Index 1–Index 4* in Table 3.

For each subsystem in the MECADNs resilience assessment system, static and dynamic indicators are defined based on their evaluation dimensions, time dependence, and data sources. Static indicators (SIs) focus on the inherent network structure (independent of time and

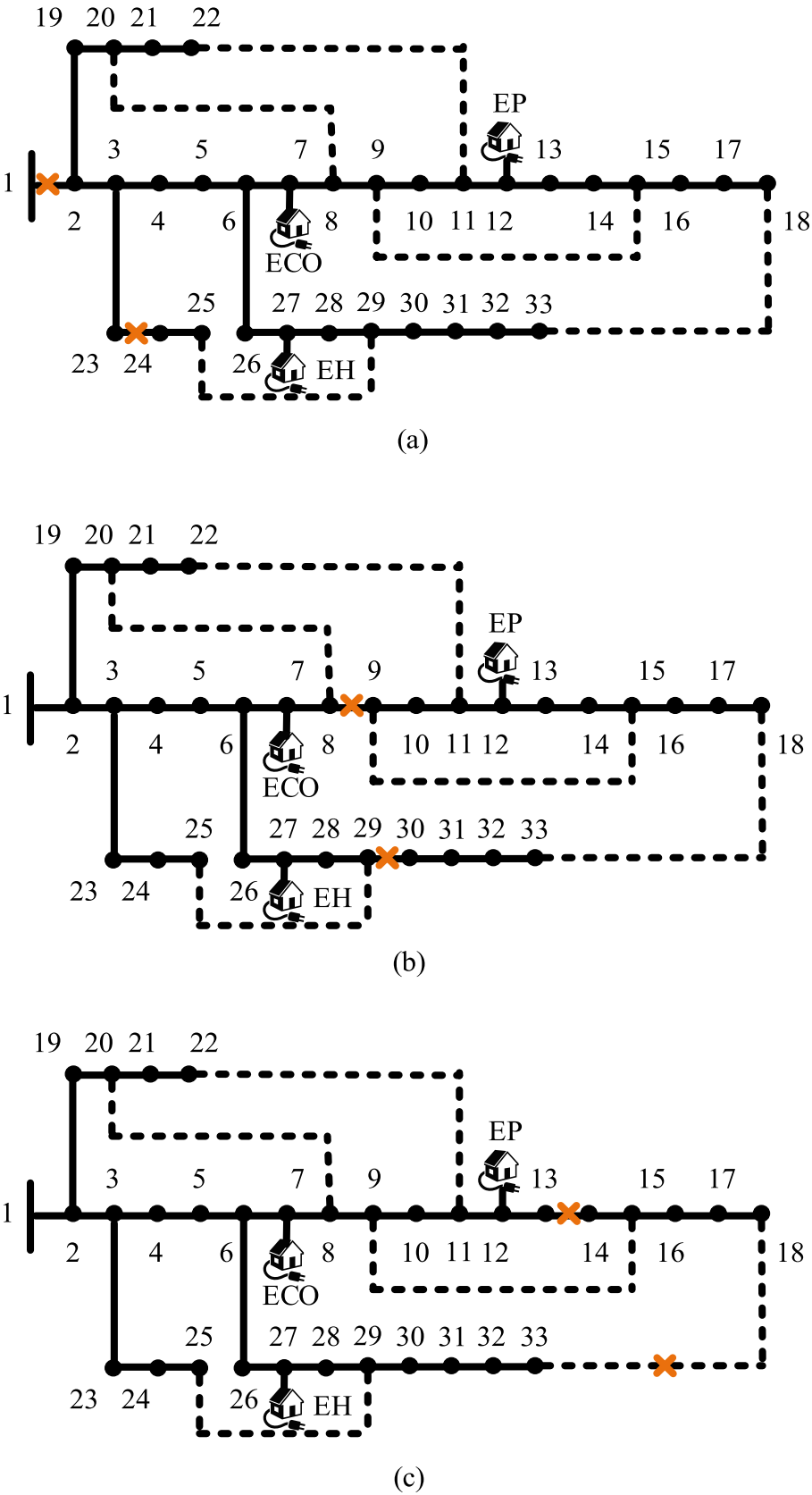


Fig. 7. Fault evolution path considering the multi-period typhoon fault scenarios. (a) Stage 1, (b) stage 2, and (c) stage 3.

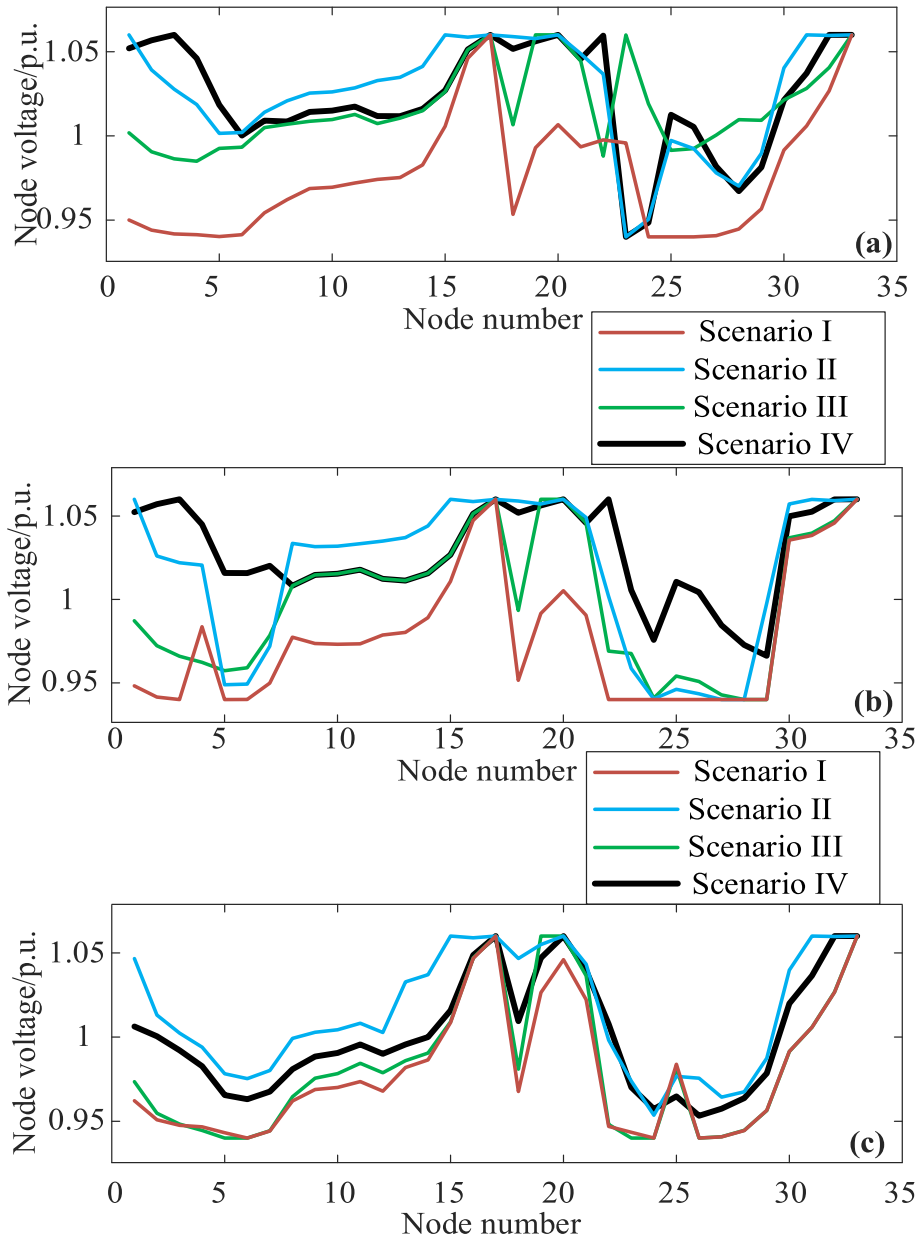


Fig. 8. Node voltage of the ADNs in the MECADNs considering the multi-period typhoon fault scenarios. (a) Stage 1, (b) stage 2, and (c) stage 3.

extreme-event processes) and are analyzed using graph theory. They assess the subsystem's intrinsic resilience potential by quantifying structural characteristics such as branch redundancy, resource redundancy, source-load power ratio, and network complexity—reflecting the system's ability to withstand extreme events through its inherent structural design. Dynamic indicators (DIs) focus on the time-varying performance of the subsystem during and after typhoon events and are derived from the resilience trapezoid theory. These indicators are mostly constructed from mathematical relationships among physical quantities (e.g., power, temperature, pressure, water flow) and require operational data from the MECADNs. They are calculated using power flow/thermal flow/hydraulic flow section data across various time periods, thereby capturing the dynamic evolution of system functions.

Table 3 reveals coordinated resilience performance across all subsystems. In the ADNs, SI ranges from 0.0015 to 0.1939, and DI ranges from 0.1288 to 0.9720, reflecting balanced static-dynamic resilience. In the NGS, SI spans 0.0139–0.7778, and DI spans 0.0968–1, indicating robust fault tolerance for gas supply. In the WN, SI ranges from 0.0010 to

0.2002, and DI ranges from 0.0061 to 1, indicating stable water network operation under fault conditions. In the DHS, SI spans 0.0042–0.1875, and DI spans 0.0037–1, showing reliable heat supply resilience. In addition, this work quantifies the system resilience score under different scenarios, where Scenario I scores 42.00, Scenario II scores 43.39, Scenario III scores 43.08, and Scenario IV achieves the highest score of 43.42. This quantitative comparison, coupled with the qualitative consistency of the detailed indicators in Table 2, confirms that Scenario IV (the proposed method) optimizes the BESDs configuration to best improve the MECADNs' resilience under fault conditions.

4.2. MECADNs model based on 118-node system

To validate the scalability of the proposed method in large-scale multi-energy systems, a MECADNs model is constructed based on the modified IEEE 118-node ADNs [63], as illustrated in Fig. 11.

This model integrates a 10-node NGS, a 35-node DHS, and a 50-node water network, with inter-system coupling nodes distinguished by color

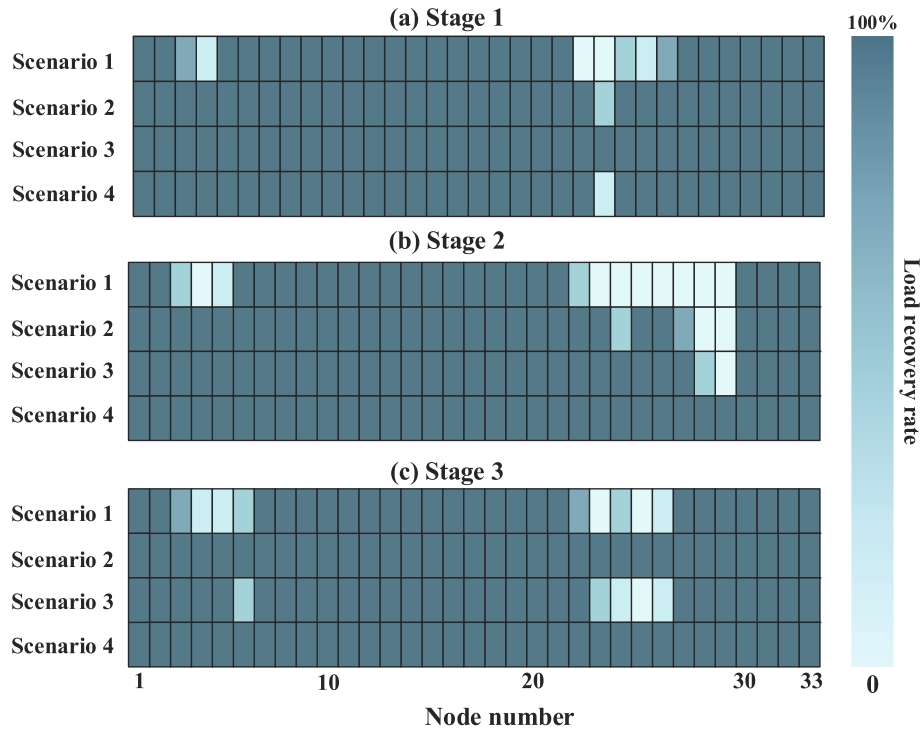


Fig. 9. Fault recovery data of the ADNs in the MECADNs considering the multi-period typhoon fault scenarios. (a) Stage 1, (b) stage 2, and (c) stage 3.

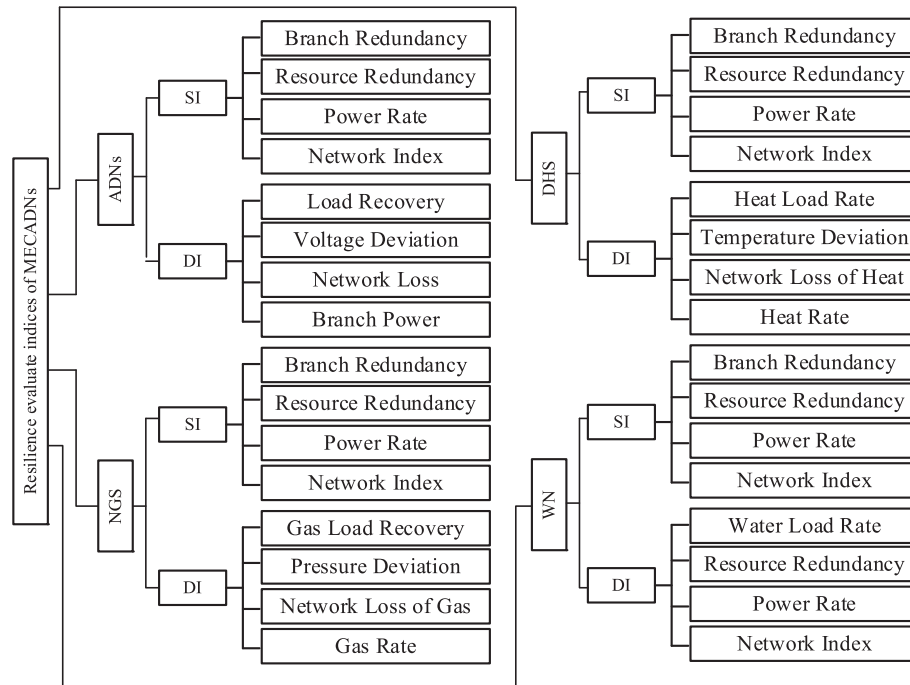


Fig. 10. Resilience indexes of the MECADNs.

coding—yellow nodes (e.g., node 2) are coupled to the NGS, cyan nodes (e.g., node 7) to the WN, and red nodes (e.g., node 15) to the DHS.

To characterize fault behaviors in large-scale MECADNs under typhoon events, the OPA model is extended to generate typical fault scenarios by incorporating the spatiotemporal variation of typhoon impacts, consistent with the 33-node system's scenario generation logic but scaled to match the 118-node system's higher component count. Initially, 800 groups of fault scenarios are generated to cover failures that reflect the system's complexity. Through information entropy

quantification, three representative fault scenarios are determined: (1) faults occur in the ADNs branches 13–14 & 78–79, information entropy $E = 12,415.14$; (2) faults occur in the ADNs branches 29–30 & 104–105, information entropy $E = 12,414.57$; (3) faults occur in the ADNs branches 44–45 & 75–76, information entropy $E = 12,415.51$, which serve as benchmark disturbance conditions for optimization configuration of the BESDs.

Fig. 12 compares the convergence of three algorithms (PSO, SAO, and the SAO-PSO hybrid algorithm) in solving the BESDs optimization

Table 3
Resilience indices in the fault state of Scenario IV.

Item	Index	Index 1	Index 2	Index 3	Index 4
ADNs	SI	0.0251	0.0015	0.0161	0.1939
	DI	0.9720	0.0111	0.1288	0.7352
NGS	SI	0.2222	0.0139	0.2375	0.7778
	DI	1	0.1381	0.4041	0.0968
WN	SI	0.0504	0.0010	0.0011	0.2002
	DI	1	0.0793	0.1055	0.0061
DHS	SI	0.1875	0.0042	0.0177	0.1853
	DI	1	0.0734	0.1241	0.0037

model (with the inner-layer model uniformly solved via CPLEX + YALMIP), and key insights can be drawn: the PSO algorithm converges rapidly within 5 iterations but succumbs to local optimality, stabilizing at an objective function value of ~ 1.8 – 1.9 , which highlights the PSO's limitation of premature convergence in large-scale problems where the complex solution space leads to suboptimal results; the SAO algorithm demonstrates slow initial convergence (requiring ~ 15 iterations to stabilize) yet avoids local traps, converging to an objective function value of ~ 1.2 – 1.3 , and while it ensures global search capability, the slow convergence rate reduces efficiency in large systems; in contrast, the SAO-PSO hybrid algorithm achieves a balance between speed and globality by inheriting the PSO's rapid early-stage convergence (stabilizing around the 10th iteration) and the SAO's robust global search, ultimately converging to a lower objective function value (near 1.1)—this verifies the hybrid algorithm's superiority in both convergence speed and solution accuracy for large-scale MECADNs and addresses the

trade-off between search efficiency and global optimality.

Table 4 presents the optimal configuration results of the BESDs in the MECADNs based on a 118-node system. Comparing Table 4 with Table 2 reveals consistent algorithm performance trends across scales (SAO-PSO > SAO > PSO), alongside scale-dependent differences in absolute costs and capacity magnitudes.

To further validate the proposed method's scalability, we extend the comparison to the 118-node system (with a total active load of 22,709 MW) by defining a resilience-oriented rule-of-thumb baseline. This baseline follows the engineering practices: it is designed to meet 80 % of the system's power load for 12 h with a 1.3 safety margin. The required total BESD capacity for this baseline is about $22,709 \text{ MW} \times 80 \% \times 12 \text{ h} \times 1.3 \approx 283.41 \text{ MWh}$. In stark contrast, the proposed SAO-PSO hybrid algorithm achieves a total BESD capacity of 74.87 MWh. This represents a remarkable 73.6 % reduction in capacity compared to the baseline. This comparison further underscores the framework's superiority in delivering cost-effective, resilience-optimized BESD configurations for large-scale MECADNs. In addition, the hybrid algorithm's relative cost savings compared to the PSO are smaller, mainly due to increased system complexity. Still, the hybrid algorithm remains the best choice while highlighting the challenges of optimizing complex multi-energy systems.

To illustrate the impact of the inner-layer model on the operation of water, gas, and heat subsystems in large-scale MECADNs, Fig. 13 presents comparative results with and without the inner-layer model.

For the water system in Fig. 13(a), the water load curve without the inner-layer model fluctuates, with peak values reaching around $1 \times 10^4 \text{ L}$ at certain nodes; in contrast, the curve with the inner-layer model

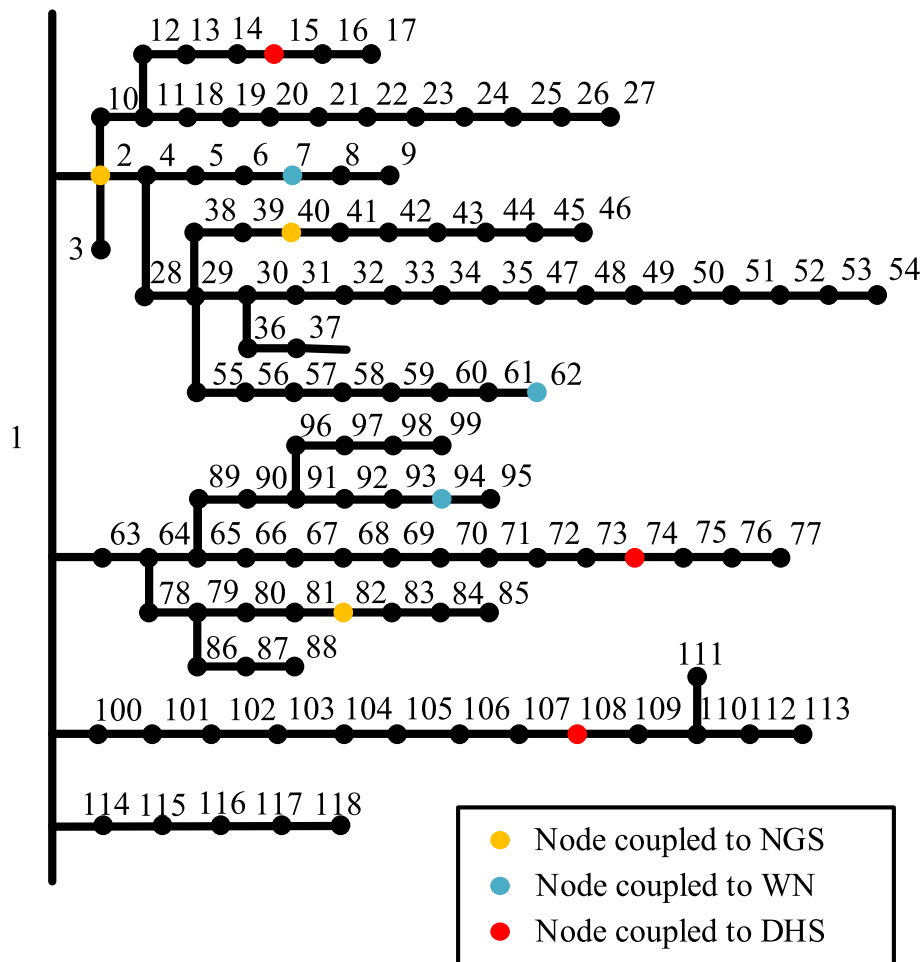


Fig. 11. MECADNs model based on the modified IEEE 118-node system.

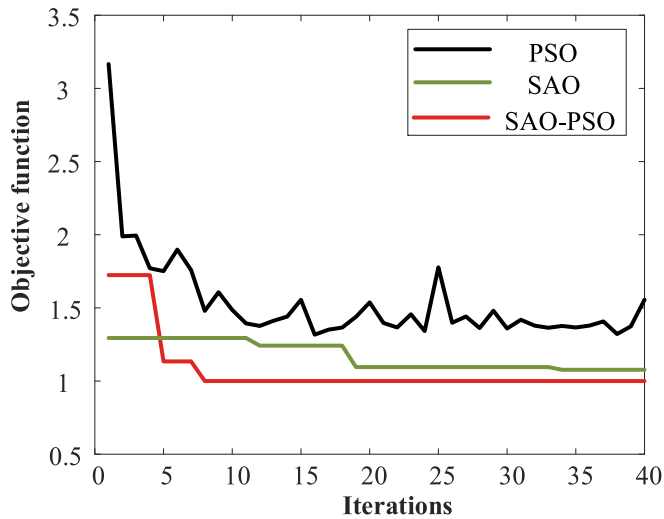


Fig. 12. Performance comparison of different algorithms to solve the proposed model in the large-scale MECADNs.

Table 4

Optimal configuration results of the BESDs in the MECADNs model based on a 118-node system.

Case	Site	Capacity/MWh	Cost/k¥	Total cost/k¥
1	33	21.045	14,941.95	58,554.41
	38	30.129	21,391.59	
	67	31.297	22,220.87	
2	31	18.132	12,873.72	58,332.18
	45	34.699	24,636.29	
	97	29.327	20,822.17	
3	37	24.011	17,047.81	53,155.57
	55	15.967	11,336.57	
	112	34.889	24,771.19	

becomes smoother, indicating improved water demand distribution across nodes. Turning to the heat system in Fig. 13(b), the heat load without the inner-layer model exhibits intense oscillations, while the curve with the inner-layer model shows reduced fluctuations and a more balanced profile, with peak load decreasing from ~0.15 MW to below 0.12 MW at key nodes. For the gas system in Fig. 13(c), the gas load

without the inner-layer model exhibits erratic peaks, whereas the load with the model is more effectively regulated. Regarding the gas pressure in Fig. 13(d), the pressure curve without the inner-layer model drops sharply; in contrast, the curve with the inner-layer model remains more stable, staying above approximately 0.95 p.u. at most nodes. These results demonstrate that the inner-layer model enables coordinated optimization of water-gas-heat operations, verifying the model's scalability across different system scales.

In addition, the multi-period typhoon fault scenarios are considered in large-scale MECADNs, and the OPA model is used to determine the fault evolution path in Fig. 14. There are four scenarios designed below:

Scenario I: No BESDs configuration.

Scenario II: BESDs optimized by the PSO algorithm.

Scenario III: BESDs optimized by the SAO algorithm.

Scenario IV: BESDs optimized by the SAO-PSO hybrid algorithm.

To verify the resilience enhancement of BESDs configuration in large-scale MECADNs under multi-period typhoon faults, Fig. 15 presents load recovery rate heatmaps for three stages across four scenarios. Quantitatively, the average load recovery rates across all three stages are 78.3 % (Scenario I), 92.5 % (Scenario II), 94.1 % (Scenario III), and 97.8 % (Scenario IV)—a clear hierarchy that confirms Scenario IV's dominant recovery performance. In Stage 1, Scenario I shows extensive light patches (indicating low recovery), while Scenario IV maintains a recovery rate of approximately 96.2 %; in Stage 2, Scenario IV achieves near-100 % recovery for most nodes; and in Stage 3, only minor light areas exist (recovery rate about 98.5 %).

Table 5 quantifies the resilience indices for Scenario IV (the proposed method) across subsystems (ADNs, NGS, WN, DHS), distinguishing between SI and DI.

In the ADNs, SI ranges from 0.0072 to 0.2215 and DI from 0.0019 to 0.9101, reflecting balanced static-dynamic resilience. In the NGS, SI spans 0.0207–0.2839 and DI 0.0601–0.9989, demonstrating robust fault tolerance for gas supply. For the WN, SI ranges 0.0089–0.2940 and DI 0.0003–0.9891, indicating stable water network operation under faults. In the DHS, SI spans 0.1960–0.4856 and DI 0.0035–0.9917, showing reliable resilience for heat supply.

Notably, large-scale MECADNs exhibit more significant quantitative resilience improvement compared to small-scale MECADNs based on the 33-node system. For small-scale MECADNs, the resilience score rises from 42.00 (Scenario I) to 43.42 (Scenario IV, a 1.42-point increase), with load recovery rate improving from 88.6 % to 99.3 % (a 10.7-percentage-point increase). In contrast, for large-scale MECADNs based

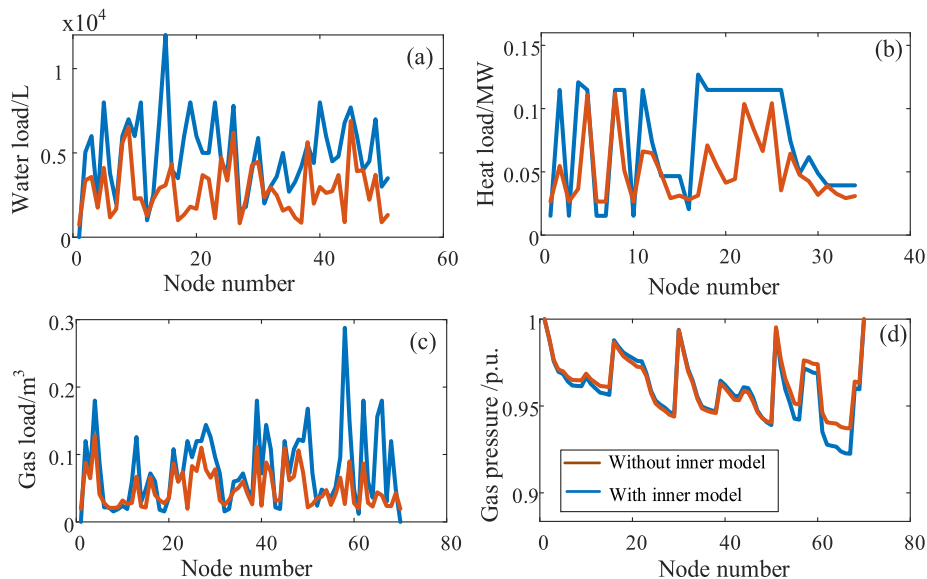


Fig. 13. The influence of the inner-layer model of BESDs configuration on the water-gas-heat systems in large-scale MECADNs.

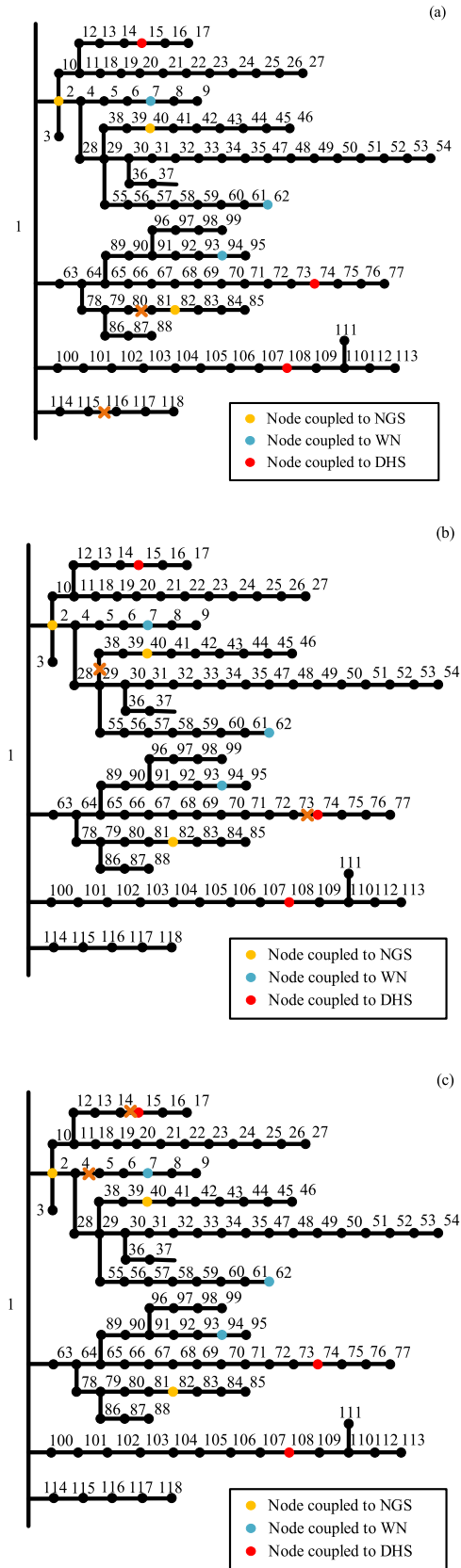


Fig. 14. Fault evolution path for large-scale MECADNs. (a) Stage 1, (b) stage 2, and (c) stage 3.

on the 118-node system, the resilience score jumps from 58.52 (Scenario I) to 60.51 (Scenario IV, a 1.99-point increase), and the load recovery rate surges from 78.3 % to 97.8 % (a 19.5-percentage-point increase). This indicates that the proposed method's resilience enhancement is more significant in large-scale systems, likely driven by three mechanisms arising from the interaction between network topology, system complexity, and algorithmic flexibility: (1) The larger, more complex multi-energy network (e.g., 118-node system) has greater topological redundancy and more optimization options compared to smaller systems, with more nodes, multi-energy links, and diverse subsystems (electricity, gas, heat), offering richer BESD configurations and broader opportunities for cross-subsystem coordination. (2) Large-scale systems demonstrate better fault propagation mitigation and buffer capacity, while distributed BESDs can more effectively prioritize critical load restoration—a capability limited in small-scale networks with fewer alternative paths. (3) The SAO-PSO hybrid algorithm is inherently suited for high-dimensional optimization in large-scale MECADNs, as its balance of the SAO and PSO can fully utilize the increased decision variables and complex constraints, whereas small-scale systems' low-dimensional optimization space limits this potential. These factors collectively enhance the method's ability to leverage coordinated cross-subsystem operation via optimizing BESDs, resulting in greater resilience gains in large-scale MECADNs.

5. Conclusions

This paper puts forward an optimal configuration method for battery energy storage devices (BESDs) to enhance the resilience of multi-energy coupled active distribution networks (MECADNs), which integrates the optimal operation of multi-energy subsystems. A two-layer optimization model is established, and a solving strategy based on the SAO-PSO hybrid algorithm is developed. The proposed approach is verified via cross-verification in small- and large-scale MECADNs based on modified IEEE 33-node, 118-node systems, with theoretical analysis, numerical simulation, and resilience index quantification combined. The main conclusions of this study are as follows:

- (1) The proposed two-layer optimization model exhibits excellent scalability across MECADNs of varying scales, and the SAO-PSO hybrid algorithm outperforms single algorithms in both solving speed and accuracy. Economically, it reduces the BESDs configuration cost by 16,287.4 k¥ in the small-scale MECADNs based on the 33-node system and by 5398.84 k¥ in the large-scale MECADNs based on the 118-node system, verifying its ability to balance global optimization and computational efficiency.
- (2) Under typical typhoon fault scenarios considering multi-stage evolution, the proposed method realizes simultaneous optimization of BESDs configuration and MECADNs operation, effectively mitigating fault impacts on multi-energy subsystems. In the small-scale MECADNs, the hybrid algorithm's configuration achieves an average load recovery rate of 99.3 % across fault stages, compared with 88.6 % for the scenario without BESDs and 98.4 % for the PSO-based configuration. In the large-scale MECADNs, the load recovery rate increases from 78.3 % (no BESDs) to 97.8 % (hybrid algorithm), with nearly 100 % recovery in the late fault stage. This performance stems from the method's ability to coordinate electricity, gas, water, and heat subsystems, smoothing load fluctuations and stabilizing state parameters.
- (3) The proposed method significantly enhances MECADNs' resilience. In terms of resilience scores, the small-scale MECADNs rise from 42.00 (no BESDs) to 43.42 (hybrid algorithm), while the large-scale MECADNs jump from 58.52 to 60.51—a 1.99-point increase that is 40.1 % higher than the small-scale MECADNs' 1.42-point gain. Subsystem resilience indices (static indicator SI: 0.0072–0.4856; dynamic indicator DI: 0.0003–0.9989) further confirm balanced fault tolerance across all energy networks.

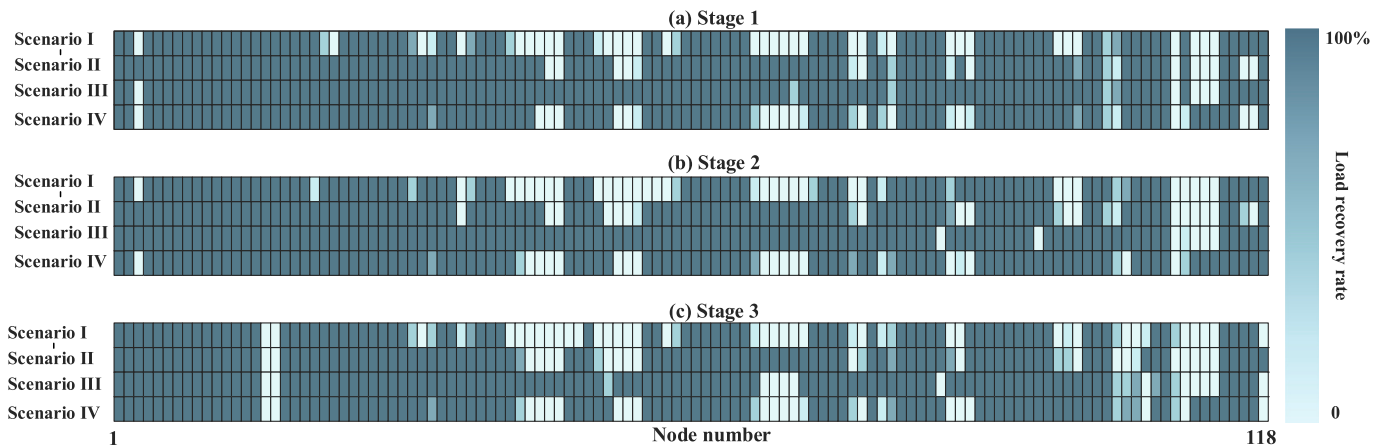


Fig. 15. Fault recovery data of the ADNs in large-scale MECADNs considering the multi-period typhoon fault scenarios. (a) Stage 1, (b) stage 2, and (c) stage 3.

Table 5

Resilience indices in the fault state of Scenario IV for large-scale MECADNs.

Item	Index	Index 1	Index 2	Index 3	Index 4
ADNs	SI	0.0823	0.0072	0.0530	0.2215
	DI	0.9101	0.0283	0.0019	0.1873
NGS	SI	0.2839	0.0207	0.0781	0.2171
	DI	0.9989	0.0601	0.0776	0.3549
WN	SI	0.2940	0.0089	0.0860	0.1941
	DI	0.9891	0.0268	0.0858	0.0003
DHS	SI	0.1960	0.3921	0.4856	0.1961
	DI	0.9917	0.0531	0.0492	0.0035

The strengths of this work lie in its integration of BESDs configuration with multi-energy operation optimization for synergistic resilience improvement, the SAO-PSO hybrid algorithm's balanced optimization accuracy and computational efficiency adaptable to different-scale MECADNs, its alignment with engineering practice via consideration of multi-stage typhoon scenarios and static-dynamic resilience indices, and its good scalability and generalizability validated across 33-node and 118-node systems. The limitations of the work include focusing on single typhoon events rather than compound disaster scenarios, omitting the CHP's feasible operating region (FOR) and nonlinear cost, which are key characteristics emphasized in [64,65], verification for ultra-large-scale MECADNs, and the lack of integration with market mechanisms to explore BESDs' revenue potential in auxiliary services.

In future work, we will (i) extend the model to compound disaster scenarios (e.g., typhoon & equipment aging) and incorporate the CHP's FOR and nonlinear cost to align with engineering practice; (ii) study scalability by analyzing ultra-large-scale MECADNs and quantify interactions between multiple BESDs clusters; (iii) explore market-oriented application paths, such as enabling configured BESDs to participate in auxiliary services and optimizing revenue models under spot market mechanisms. Related findings will be reported in later articles.

CRediT authorship contribution statement

Lei Chen: Writing – original draft, Validation, Supervision, Methodology, Data curation, Conceptualization. **Yi Zheng:** Writing – original draft, Validation, Software, Data curation. **Bude Su:** Writing – review & editing, Validation, Software, Data curation. **Musi Li:** Writing – review & editing, Software, Formal analysis, Data curation. **Yifei Li:** Writing – review & editing, Software, Data curation. **Hongkun Chen:** Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Smart Grid-National Science and Technology Major Project under Grant 2024ZD0800600.

Data availability

No data was used for the research described in the article.

References

- [1] C. Wang, C. Liu, X. Zhou, Y. Li, G. Zhang, Hierarchical optimal dispatch of active distribution networks considering flexibility auxiliary service of multi-community integrated energy systems, *IEEE Trans. Ind. Appl.* 61 (2) (2025) 2770–2781.
- [2] J. Yan, J. Zhang, L. Zhang, C. Deng, J. Zhang, X. Wang, T. Gao, Enhanced resilience and efficiency in multi-energy systems via stochastic gradient-driven robust optimization, *Prot. Control Mod. Power Syst.* (2025), <https://doi.org/10.23919/PCMP.2024.000412>.
- [3] Y. Zou, Y. Xu, J. Li, Aggregator-network coordinated peer-to-peer multi-energy trading via adaptive robust stochastic optimization, *IEEE Trans. Power Syst.* 39 (6) (2024) 7124–7137.
- [4] D. Xu, J. Xu, C. Qian, Z. Wu, Q. Hu, A pseudo-measurement modelling strategy for active distribution networks considering uncertainty of DGs, *Prot. Control Mod. Power Syst.* 9 (5) (2024) 1–15.
- [5] S.R. Seyednouria, A. Safari, S.N. Ravadanegh, N. Taghizadegan, A. Quteishat, Distributed and adaptive robust energy management of an active distribution network with multi-microgrid considering transactive energy, *Electr. Pow. Syst. Res.* 248 (2025) 111969.
- [6] B.-C. Oh, Y.-G. Son, D. Zhao, C. Singh, S.-Y. Kim, A two layer approach for networked microgrid planning considering multiple contingencies and resilience, *IEEE Trans. Power Syst.* 39 (4) (2024) 5620–5630.
- [7] Z. Wang, H. Hou, R. Wei, Z. Li, A distributed market-aided restoration approach of multi-energy distribution systems considering comprehensive uncertainties from typhoon disaster, *IEEE Trans. Smart Grid* 16 (5) (2025) 3743–3757.
- [8] J. Wu, J. Gu, S. Liu, Z. Jin, Strategies for improving resilience of regional integrated energy systems in the prevention-resistance phase of integration, *Prot. Control Mod. Power Syst.* 8 (2) (2023) 1–18.
- [9] Y. Gao, Z. Zhang, K. Meng, W. Liu, R. Gao, Graph reinforcement learning for real-time dynamic reconfiguration and fault management in energy storage networks, *J. Energy Storage* 125 (2025) 116833.
- [10] P. Yu, X. Xiong, X. He, J. Zhu, J. Liang, D. Nan, Emergency load shedding strategy with warning and delay time based on energy storage, *Prot. Control Mod. Power Syst.* 10 (4) (2025) 28–41.
- [11] Y. Wang, D. Qiu, X. Sun, Z. Bie, G. Strbac, Coordinating multi-energy microgrids for integrated energy system resilience: a multi-task learning approach, *IEEE Trans. Sustain. Energy* 15 (2) (2024) 920–937.
- [12] T. Li, X. Han, H. Sun, W. Wu, D. Zhang, A multi-agent distributed operation solution to electricity and natural gas distribution systems for enhancing resilience, *IEEE Trans. Smart Grid* 15 (2) (2024) 1525–1543.

- [13] Y. Yan, Y. Chen, Z. Cui, T. Xiao, Probabilistic resilience assessment of urban distribution power grids by fast inference of multi-source multi-terminal network reliability, *Reliab. Eng. Syst. Saf.* 261 (2025) 111077.
- [14] Z. Bai, D. Xu, B. Zhou, X. Yang, H. Yang, J. Liu, Computing optimal resilience-oriented operation of distribution systems considering heterogenous consumer-side microgrids, *IEEE Trans. Consum. Electron.* 71 (2) (2025) 5866–5875.
- [15] X. Jing, W. Qin, P. Wang, X. Han, H. Yao, Z. Zhu, Resilience-promoting decentralized robust synthetic restoration strategy for regional integrated energy system, *CSEE J. Power Energy Syst.* (2025), <https://doi.org/10.17775/CSEEJPES.2022.06610>.
- [16] J. Zhang, Y. Li, H. Yuan, G. Du, M. Zhang, A demand-based three-stage seismic resilience assessment and multi-objective optimization method of community water distribution networks, *Reliab. Eng. Syst. Saf.* 250 (2024) 110279.
- [17] N. Mohammadali, M.-I. Behnam, A. Jamshid, A. Mohammadali, G. Reza, Three-layer resilience-aware scheduling framework in a hydrogen-integrated distribution network considering fuel cell electric vehicles, *Energy* 334 (2025) 137415.
- [18] C. Cai, F. Gan, Y. Cui, B. Li, S. Hou, Entropy-driven multi agent deep reinforcement learning for resilient distribution networks: coordinating MESS and microgrids, *Int. J. Electr. Power Energy Syst.* 170 (2025) 110968.
- [19] M.S. Javadi, C.S. Gouveia, L.M. Carvalho, A multi-temporal optimal power flow model for normal and contingent operation of microgrids, in: 2022 IEEE International Conference on Environment and Electrical Engineering and 2022 IEEE Industrial and Commercial Power Systems Europe (IEEEIC/1&CPS Europe), Prague, Czech Republic, 2022, pp. 1–6.
- [20] C. Wang, C. Cheng, T. Jin, H. Dong, Water evaporation inspired biomass-based PCM from daisy stem and paraffin for building temperature regulation, *Renew. Energy* 194 (2022) 211–219.
- [21] P. Lv, L. Liu, G. Lei, H. Dong, Pore-scale numerical investigation on the thermal storage properties of packed-bed systems with phase change material microcapsules, *J. Clean. Prod.* 405 (2023) 137071.
- [22] P. Lv, L. Liu, H. Dong, G. Lei, Y. He, Charging behavior of packed-bed thermal energy storage systems in medium and low temperature applications, *Appl. Energy* 373 (2024) 123893.
- [23] Y. Wang, Z. Su, Z. Huang, L. Shi, H. Dong, Numerical study on the efficiency improvement of household thermal energy storage device by optimized finned tube heat exchanger, *Appl. Therm. Eng.* 275 (2025) 126816.
- [24] J. Chen, W. Du, K. Cheng, X. Li, J. Guo, P. Lv, H. Dong, Design analysis of a hybrid printed circuit heat exchanger for precooling in hydrogen refueling station, *Int. Commun. Heat Mass Transf.* 163 (2025) 108765.
- [25] M.S. Javadi, Unlocking responsive flexibility within local energy communities in the presence of grid-scale batteries, *Sustain. Cities Soc.* 114 (2024) 105697.
- [26] A. Villamarín-Jacome, R. Moreno, M. Panteli, M. Noebels, E.A.M. Ceseña, R. Preece, Optimizing network expansion and battery storage to mitigate cascading outages via detailed system failures simulations, *IEEE Access* 13 (2025) 81462–81473.
- [27] Z. Ding, Z. Xu, J. Wang, G. Sun, Q. Lu, Y. Zhang, Z. Dou, X. Wei, Resilient load restoration with distributed mobile resources: a model-free approach, *IEEE Trans. Ind. Appl.* 61 (4) (2025) 5525–5536.
- [28] M.T. Ettehad, R.M. Chabani, M.T. Ameli, Improving power distribution resilience through optimal PV and BES allocation with a cost-based optimization framework for normal and emergency conditions, *IEEE Access* 13 (2025) 36436–36450.
- [29] Z. Wen, X. Zhang, G. Wang, Y. Li, J. Qiu, F. Wen, Data-driven stochastic-robust planning for resilient hydrogen-electricity system with progressive hedging decoupling, *IEEE Trans. Sustain. Energy* 16 (3) (2025) 1545–1561.
- [30] Q. Wang, X. Zhang, Y. Xu, Z. Yi, D. Xu, Planning of stationary-mobile integrated battery energy storage systems under severe convective weather, *IEEE Trans. Sustain. Energy* 16 (2) (2025) 1253–1268.
- [31] Z. Liu, Y. Zhang, Y. Ning, Emergency mobile energy storage optimal allocation in microgrid-integrated distribution networks considering economic and resilience benefits, *Energy* 322 (2025) 135633.
- [32] D. Ping, C. Li, X. Yu, Z. Liu, R. Tu, Y. Zhou, City-scale information modelling for urban energy resilience with optimal battery energy storages in Hong Kong, *Appl. Energy* 378 (2025) 124813.
- [33] H. Wang, B. Qin, S. Hong, Q. Cai, F. Li, T. Ding, H. Li, Optimal planning of hybrid hydrogen and battery energy storage for resilience enhancement using bi-layer decomposition algorithm, *J. Energy Storage* 110 (2025) 115367.
- [34] W. Ma, D. Tang, M. Dong, H. Arasteh, J.M. Guerrero, Coordinated robust configuration of soft open point and energy storage systems for resilience enhancement of integrated multi-energy system at ports, *Appl. Energy* 401 (2025) 126644.
- [35] J. Wang, A. Muammer, M.A. Rahman, A.H. Alenezi, D. Mohamed, T. Hai, Multi-objective planning and optimal configuration of wind, solar, and energy storage in interconnected microgrid clusters using Vine Copula scenario generation and antlion optimization, *Renew. Energy* 256 (2026) 124313.
- [36] B. Mousumi, Fuel constrained sustainable water-gas-heat-power generation expansion planning of isolated microgrid, *Int. J. Hydrogen Energy* 91 (2024) 1289–1305.
- [37] Q. Sun, Z. Wu, W. Gu, X. Zhang, Y. Lu, P. Liu, S. Lu, H. Qiu, Resilience assessment for integrated energy system considering gas-thermal inertia and system interdependency, *IEEE Trans. Smart Grid* 15 (2) (2024) 1509–1524.
- [38] Y. Zheng, J. Wang, S. Guan, S. Chen, Y. Zhao, Hybrid data-model driven flexible convex polyhedron construction for solving optimal electricity-heat-gas flow, *IEEE Trans. Power Syst.* 40 (4) (2025) 3295–3305.
- [39] R. Sadnan, A. Dubey, Distributed optimization using reduced network equivalents for radial power distribution systems, *IEEE Trans. Power Syst.* 36 (4) (2021) 3645–3656.
- [40] Y. He, Y. You, Z. He, H. Lu, L. Chen, Y. Jiang, A cascading fault path prediction method for integrated energy distribution networks based on the improved opa model under typhoon disasters, *Energy Eng.* 121 (10) (2024) 2825–2849.
- [41] B. Sheykhlouei, T. Abedinzadeh, L. Mohammadian, B. Mohammadi-Ivatloo, Optimal co-scheduling of distributed generation resources and natural gas network considering uncertainties, *J. Energy Storage* 21 (2019) 383–392.
- [42] L. Wang, J. Yuan, X. Qiao, X. Kong, Optimal rule based double predictive control for the management of thermal energy in a distributed clean heating system, *Renew. Energy* 215 (2023) 118924.
- [43] P. Zhao, S. Li, P. Hu, C. Gu, Z. Cao, Y. Xiang, Managing water-energy-carbon Nexus for urban areas with ambiguous moment information, *IEEE Trans. Power Syst.* 38 (5) (2023) 4432–4446.
- [44] W. Li, Y. Zou, H. Yang, X. Fu, S. Xiang, Z. Li, Two-stage stochastic energy scheduling for multi-energy rural microgrids with irrigation systems and biomass fermentation, *IEEE Trans. Smart Grid* 16 (2) (2025) 1075–1087.
- [45] T. Qin, M. Liu, S. Ji, D. Cai, Parameter weight analysis of synchronous induction electromagnetic coil launch system based on the entropy weight method, *IEEE Trans. Plasma Sci.* 52 (5) (2024) 1865–1873.
- [46] M.S. Javadi, M. Lotfi, M. Gough, A.E. Nezhad, S.F. Santos, J.P.S. Catalão, Optimal spinning reserve allocation in presence of electrical storage and renewable energy sources, in: 2019 IEEE International Conference on Environment and Electrical Engineering and 2019 IEEE Industrial and Commercial Power Systems Europe (IEEEIC/1&CPS Europe), Genova, Italy, 2019, pp. 1–6.
- [47] M. Mahdavi, M.S. Javadi, F. Wang, J.P.S. Catalão, An efficient model for accurate evaluation of consumption pattern in distribution system reconfiguration, *IEEE Trans. Ind. Appl.* 58 (3) (2022) 3102–3111.
- [48] J. Murali, T. Shankar, Improving the lifetime of UWSN using hybrid PSO-EULC algorithm, *IEEE Access* 13 (2025) 162057–162071.
- [49] Y. Wu, H. Xiao, H. Xu, H. Chen, Y. Zhang, Q. Yuan, Robust MEMS temperature control platform with GA PSO-fuzzy PID control, *IEEE Sensors J.* 25 (16) (2025) 30570–30578.
- [50] L. Deng, S. Liu, Snow ablation optimizer: a novel metaheuristic technique for numerical optimization and engineering design, *Expert Syst. Appl.* 225 (2023) 120069.
- [51] L. Chen, Y. Li, M. Yang, J. Zhu, Y. Jiang, H. Chen, Application of flux-coupling-type SFCLs in a power system considering fault current prediction based on SAO-LSTM model, *IEEE Trans. Appl. Supercond.* 34 (8) (2024) 5602206.
- [52] C. Yao, J. Cheng, K. Pan, Data-driven planning for wireless charging lanes, *IEEE Trans. Smart Grid* 16 (4) (2025) 3179–3194.
- [53] C. Lei, S. Bu, J. Zhong, Q. Chen, Q. Wang, Distribution network reconfiguration: a disjunctive convex hull approach, *IEEE Trans. Power Syst.* 38 (6) (2023) 5926–5929.
- [54] J.-O. Lee, Y.-S. Kim, J.-H. Jeon, Robust optimization method for voltage balancer planning in bipolar DC distribution systems, *IEEE Trans. Power Syst.* 39 (5) (2024) 6592–6604.
- [55] S.A. Mansouri, A. Ahmarinejad, M.S. Javadi, J.P.S. Catalão, Two-stage stochastic framework for energy hubs planning considering demand response programs, *Energy* 206 (2020) 118124.
- [56] T. Fiori, F.G. Lavacca, F. Valente, V. Eramo, Proposal and investigation of a lite time sensitive networking solution for the support of real time services in space launcher networks, *IEEE Access* 12 (2024) 10664–10680.
- [57] M.S. Javadi, C.S. Gouveia, M. L. Carvalho, R. Silva, Optimal power flow solution for distribution networks using quadratically constrained programming and McCormick relaxation technique, in: 2021 IEEE International Conference on Environment and Electrical Engineering and 2021 IEEE Industrial and Commercial Power Systems Europe (IEEEIC/1&CPS Europe), Bari, Italy, 2021, pp. 1–6.
- [58] S. Dolatabadi, M. Ghorbanian, P. Siano, N. Hatziaargyriou, An enhanced IEEE 33 bus benchmark test system for distribution system studies, *IEEE Trans. Power Syst.* 36 (3) (2021) 2565–2572.
- [59] H. Liu, C. Wang, P. Ju, Z. Xu, S. Lei, A two layer coordinated dispatch strategy for enhancing resilience of electricity-gas system considering virtual power plants, *Int. J. Electr. Power Energy Syst.* 147 (2023) 108787.
- [60] V. Srinivas, J. Wu, B. Singh, S. Mishra, Hybrid state-estimation in combined heat and electric network using SCADA and AMI measurements, *Int. J. Electr. Power Energy Syst.* 156 (2024) 109726.
- [61] J. Liu, L. Jian, W. Wang, Z. Qiu, J. Zhang, P. Dastbaz, The role of energy storage systems in resilience enhancement of health care centers with critical loads, *J. Energy Storage* 33 (2020) 102086.
- [62] Y. He, J. Wang, C. Lei, S. Ruan, J. He, Y. Jiang, L. Chen, Resilience quantified assessment of the integrated energy coupling active distribution network containing electricity-gas-water-heat, *South. Power Syst. Technol.* (2025). <https://link.cnki.net/urlid/44.1643.TK.20250616.1341.004>.
- [63] N.L. Dehghani, A. Shafieezadeh, Multi-stage resilience management of smart power distribution systems: a stochastic robust optimization model, *IEEE Trans. Smart Grid* 13 (5) (2022) 3452–3467.
- [64] M.S. Javadi, A.E. Nezhad, S. Sabramooz, Economic heat and power dispatch in modern power system harmony search algorithm versus analytical solution, *Scientia Iran.* 19 (6) (2012) 1820–1828.
- [65] M.J. Estahbanati, Hybrid probabilistic-harmony search algorithm methodology in generation scheduling problem, *J. Exp. Theor. Artif. Intell.* 26 (2) (2014) 283–296.